



UNIVERSIDAD
TORCUATO DI TELLA

Licenciatura en Tecnología Digital

Tecnología Digital VI: Inteligencia Artificial

Backpropagation a mano

Clase práctica 16

Motivación

- ¿Cómo podemos resolver manualmente una **pasada forward** y otra **backward** de *backpropagation*?
- ¿Cómo podemos verificar el cambio en los **gradientes** con PyTorch?
- ¿Cómo es la **versión generalizada** de backpropagation, basada en **matrices**?

Organización

Etapa	Inicio		Duración
	Sección 1	Sección 2	
Introducción	08:00	09:45	05'
Ejercicio a mano	08:05	09:50	60'
Ejercicio seleccionado de la guía	09:05	10:50	30'
Fin de la clase	09:35	11:20	-

Ejercicio a mano | Panorama

Backpropagation es un método de cálculo del gradiente que se puede usar en redes neuronales artificiales y, en general, para calcular gradientes de funciones.

Usado con el algoritmo de **descenso por el gradiente**, permite entrenar redes neuronales.

Hay una **forma “sistemática” o matricial** de hacer backpropagation, que resuelve el problema para una **red genérica**, independientemente de su arquitectura.

Sin embargo, primero, en esta clase, veremos un **ejercicio hecho a mano** de backpropagation, para entender **cómo funciona a nivel matemático**.

El ejemplo que veremos es para una red neuronal, pero podría servir para cualquier tipo de función compuesta a la que se le quiera calcular el gradiente.

Backpropagation - ¿Cómo funciona?

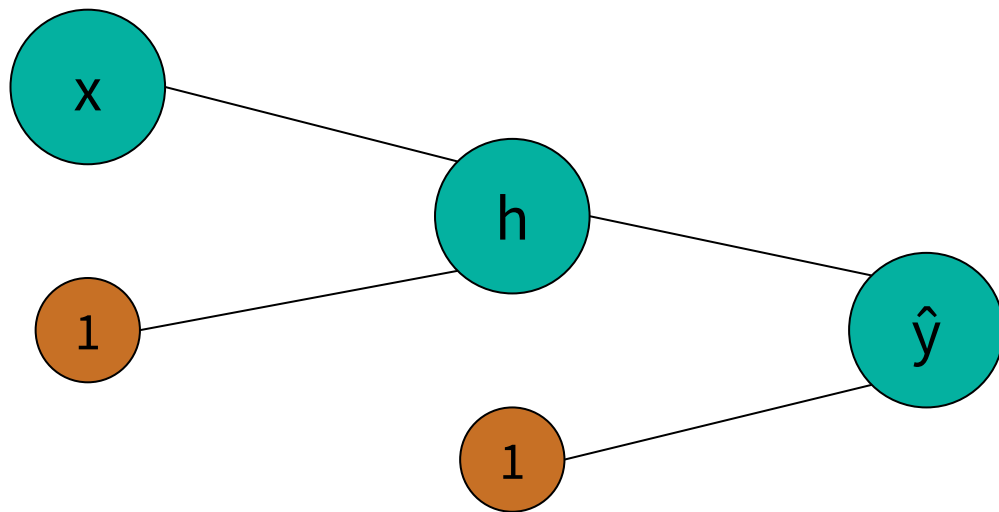
- Se hace una pasada **forward**, guardando los valores de activación de las neuronas.
- Se calculan las **derivadas** retroactivamente respecto a los pesos.
- Se modifican los pesos en base a los valores de las **derivadas** multiplicadas por un learning rate (**Gradient Descent**).
- Hay una forma “sistemática” de hacer **backpropagation**, pero en este ejercicio lo vamos a hacer a mano.

Backpropagation - Red Neuronal

Vamos a empezar con la siguiente red neuronal:

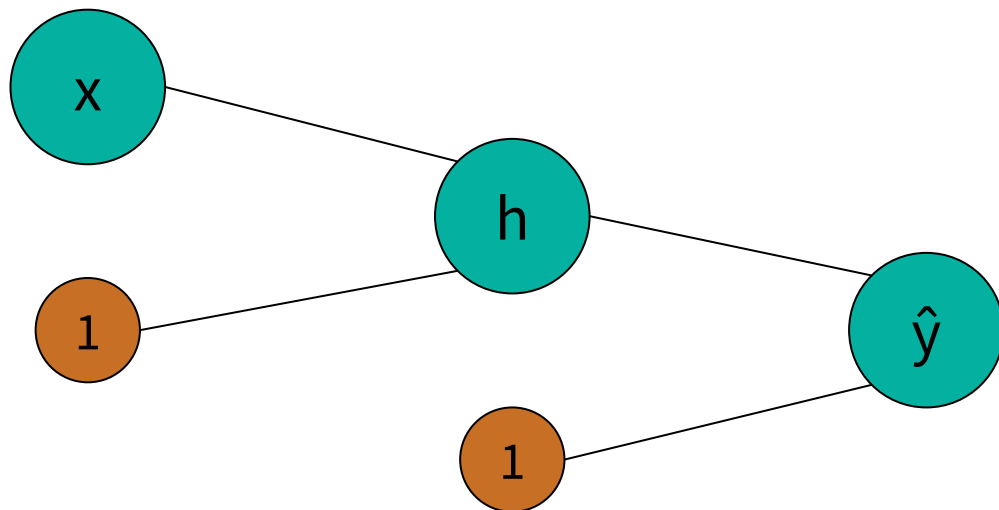
Backpropagation - Red Neuronal

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Backpropagation - Red Neuronal

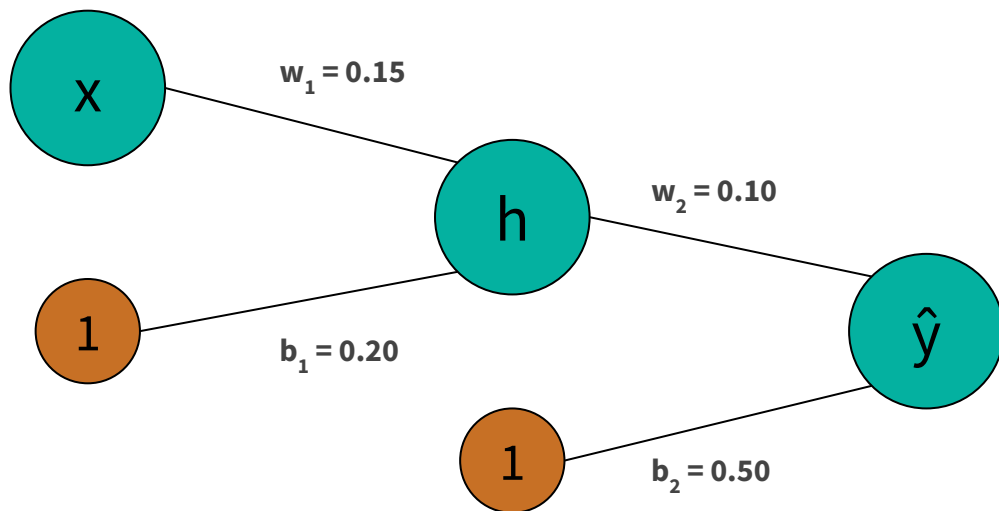
Vamos a empezar con la siguiente red neuronal:



Donde x es la neurona **input**, h una neurona de **capa oculta** y $\hat{y}$ una neurona **output**.
Los **sesgos** los representamos como un **1** constante.

Backpropagation - Red Neuronal

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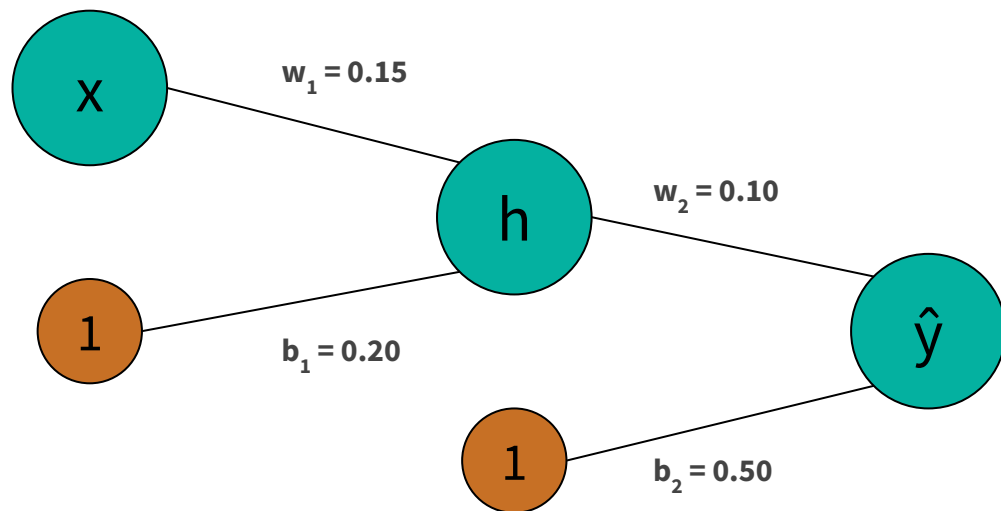
Backpropagation - Preparativos

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Vamos a mirar con más detalle a las operaciones que suceden dentro de las **neuronas**.

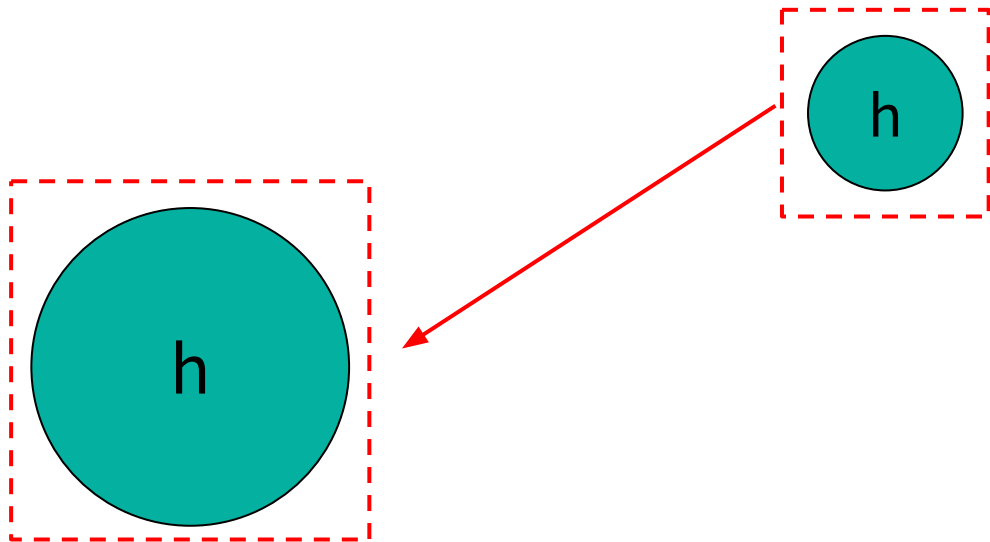
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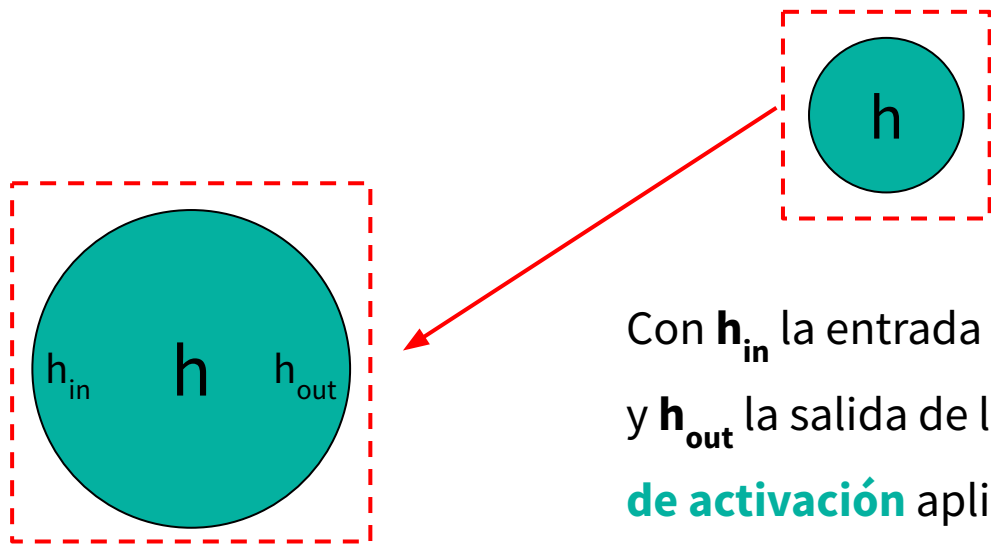
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Backpropagation - Preparativos

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Con h_{in} la entrada neta que recibe la neurona h y h_{out} la salida de la neurona h , es decir, la **función de activación** aplicada a h_{in}

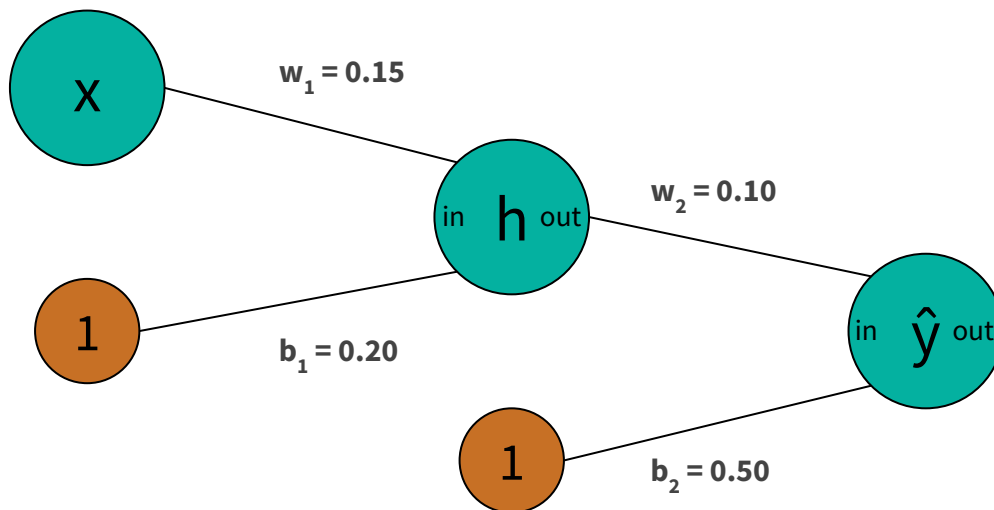
Backpropagation - Preparativos

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1$$

$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}})$$

$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$



Backpropagation - Preparativos

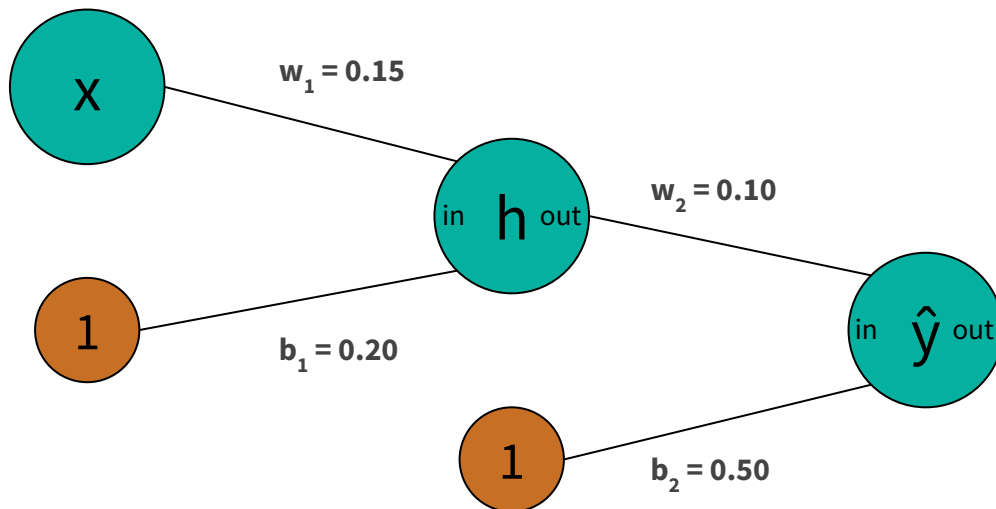
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$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}})$$

$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$\sigma(\mathbf{x})$ es la función de activación **logística**: $\sigma(x) = \frac{1}{1+e^{-x}}$



Backpropagation - Preparativos

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1$$

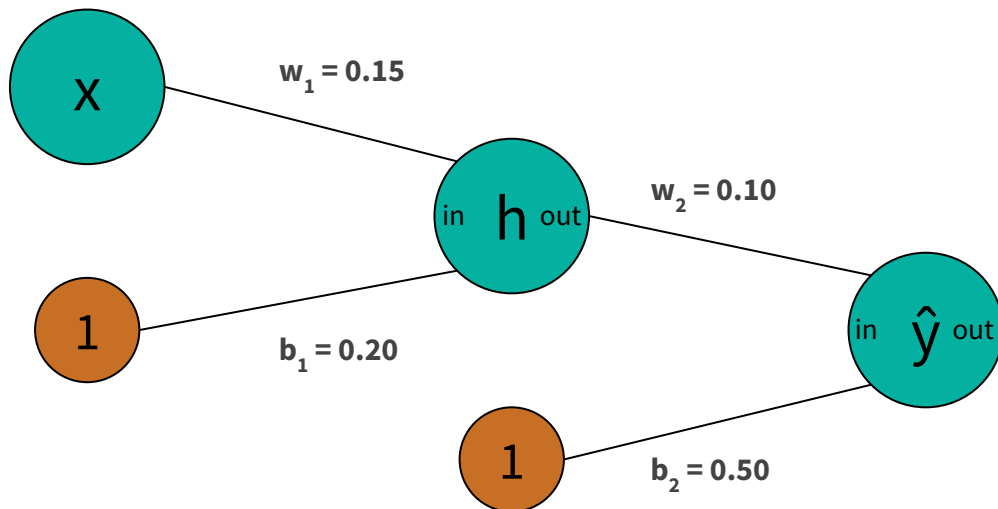
$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}})$$

$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



Backpropagation - Preparativos

Vamos a usar la **función de pérdida** $L = (y - \hat{y})^2$.

Backpropagation - Preparativos

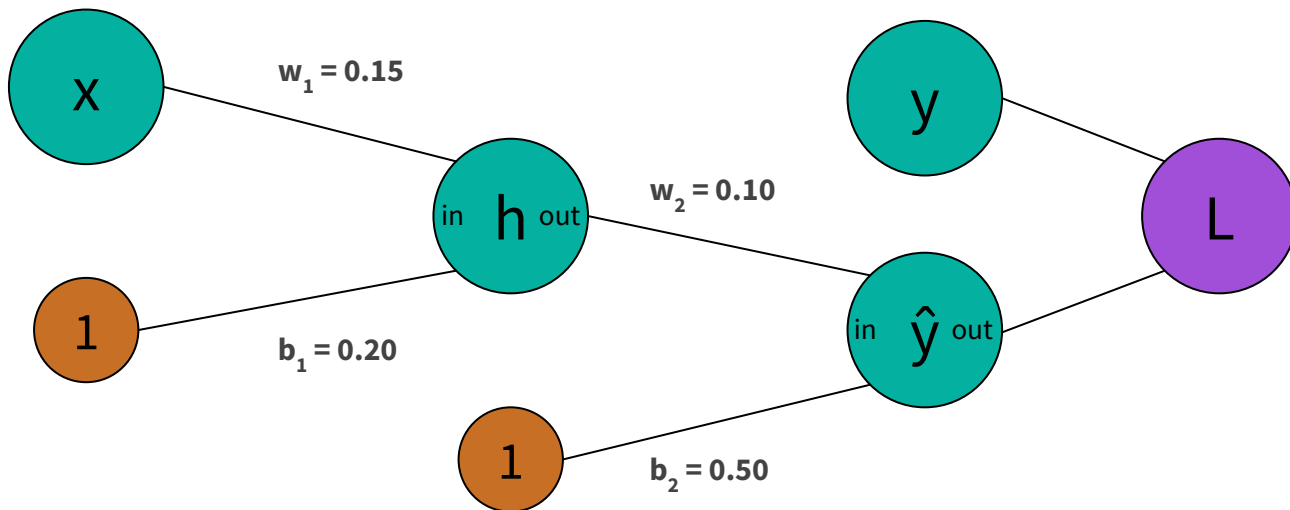
Vamos a usar la **función de pérdida** $L = (y - \hat{y})^2$.

También, vamos a agregar **L** e **y** (**salida esperada**) al grafo!

Backpropagation - Preparativos

Vamos a usar la **función de pérdida** $L = (y - \hat{y})^2$.

También, vamos a agregar **L** e **y** (**salida esperada**) al grafo!



Backpropagation - Pasada Forward

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1$$

$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}})$$

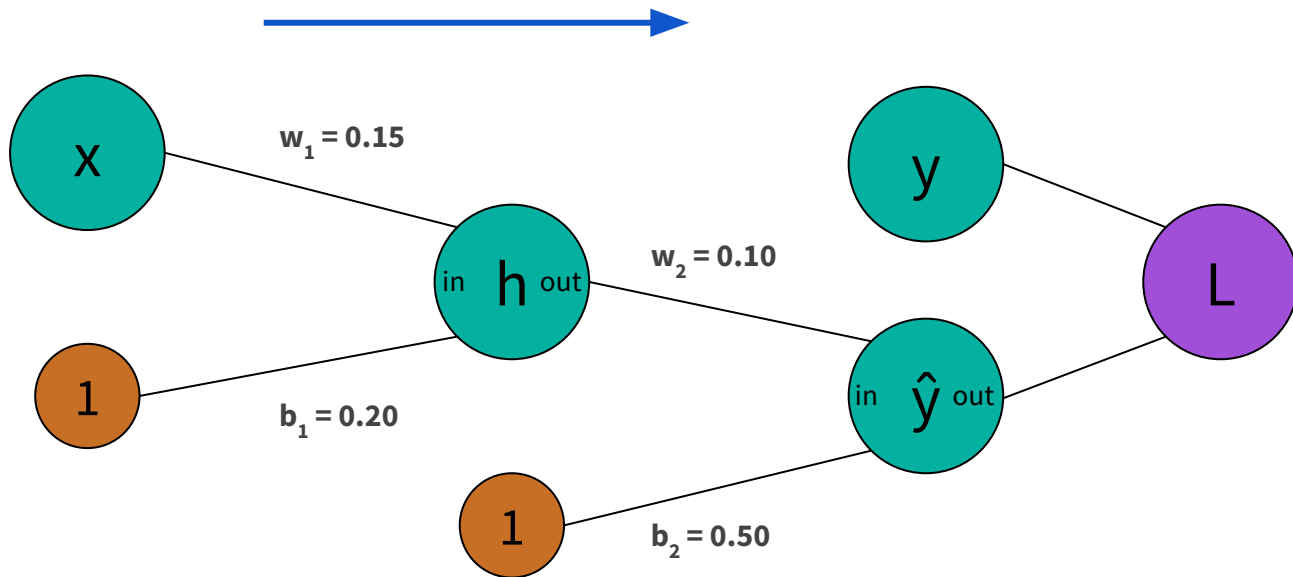
$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



Backpropagation - Pasada Forward

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1$$

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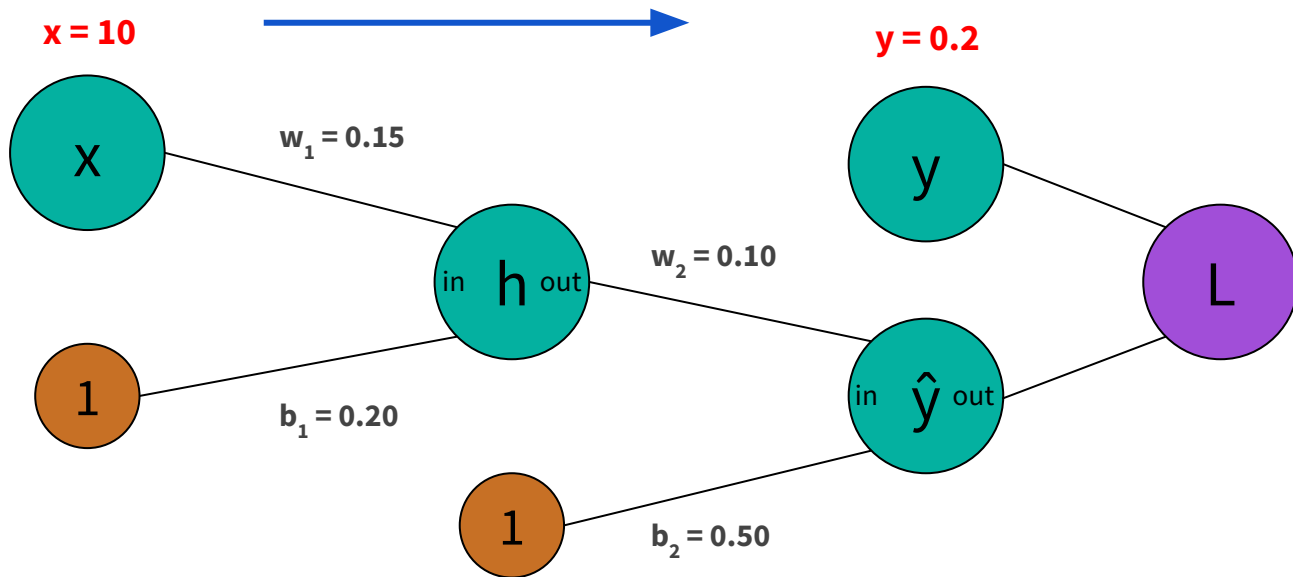
$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

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Backpropagation - Pasada Forward

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}})$$

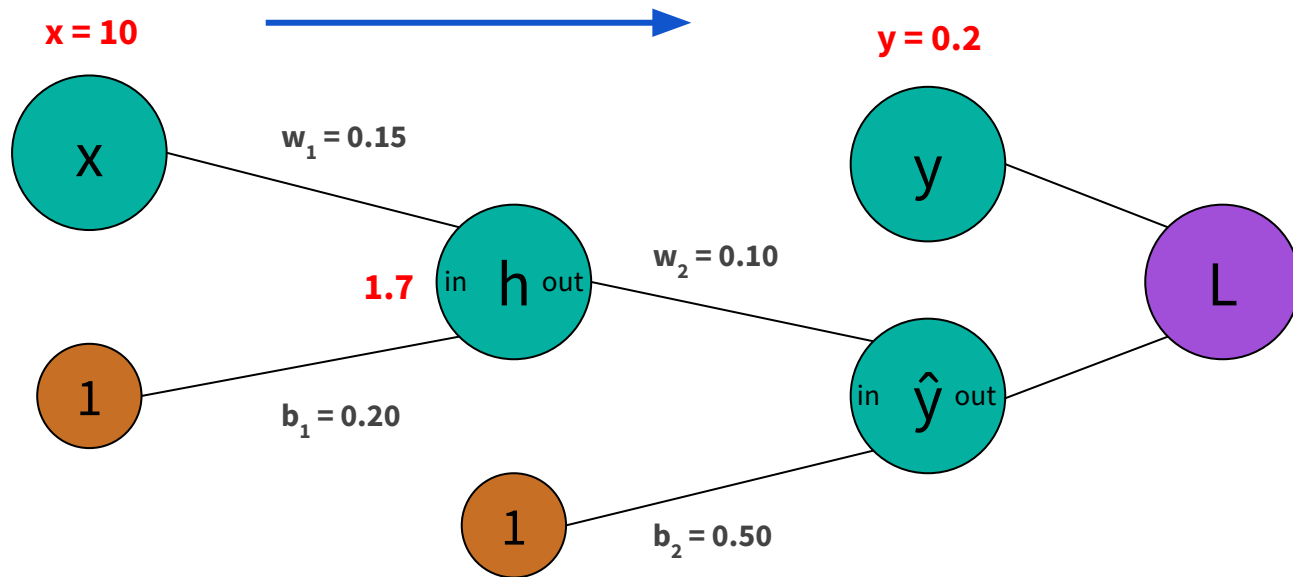
$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\mathbf{h}_{\text{in}} = 0.15 * 10 + 1 * 0.20 = 1.7$$

Backpropagation - Pasada Forward

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}}) = 0.85$$

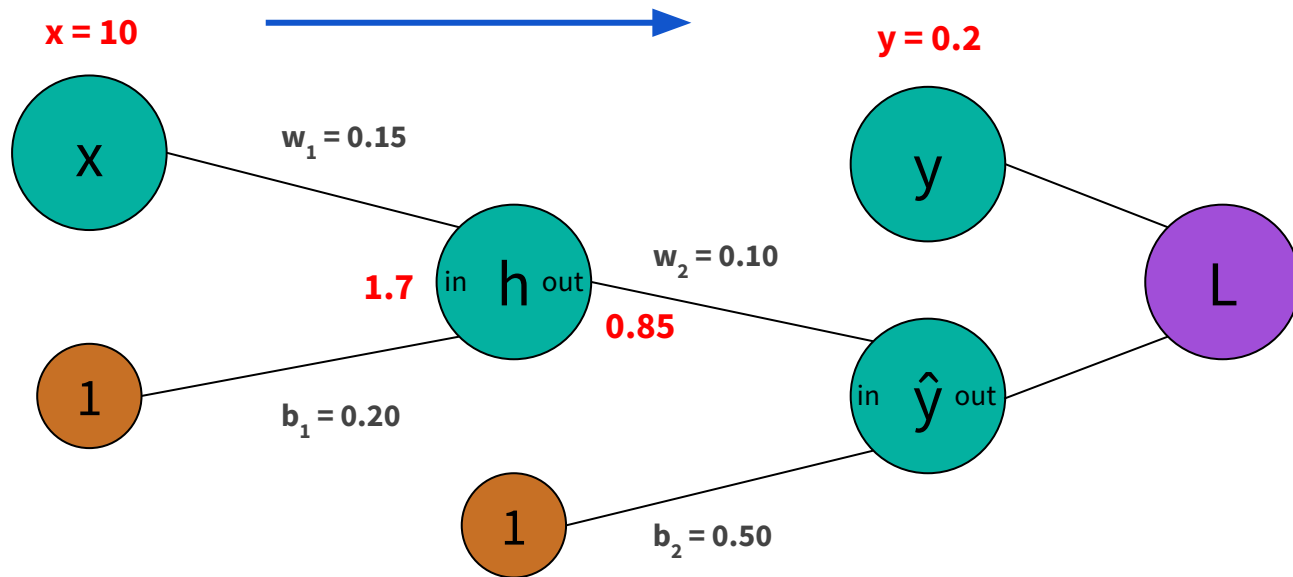
$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\mathbf{h}_{\text{in}} = 0.15 * 10 + 1 * 0.20 = 1.7$$

$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}}) = \sigma(1.7) = 0.85$$

Backpropagation - Pasada Forward

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

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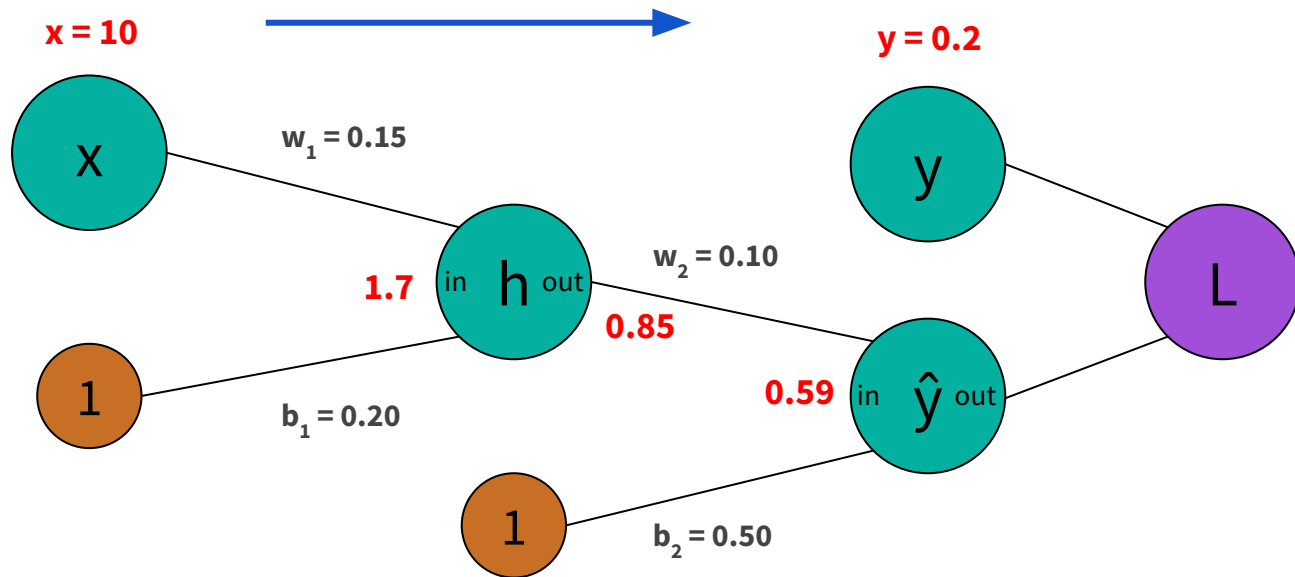
$$\hat{\mathbf{y}}_{\text{in}} = \mathbf{w}_2 \mathbf{h}_{\text{out}} + \mathbf{b}_2 = 0.59$$

$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}})$$

$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\hat{\mathbf{y}}_{\text{in}} = 0.10 * 0.85 + 1 * 0.50 = 0.59$$

Backpropagation - Pasada Forward

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

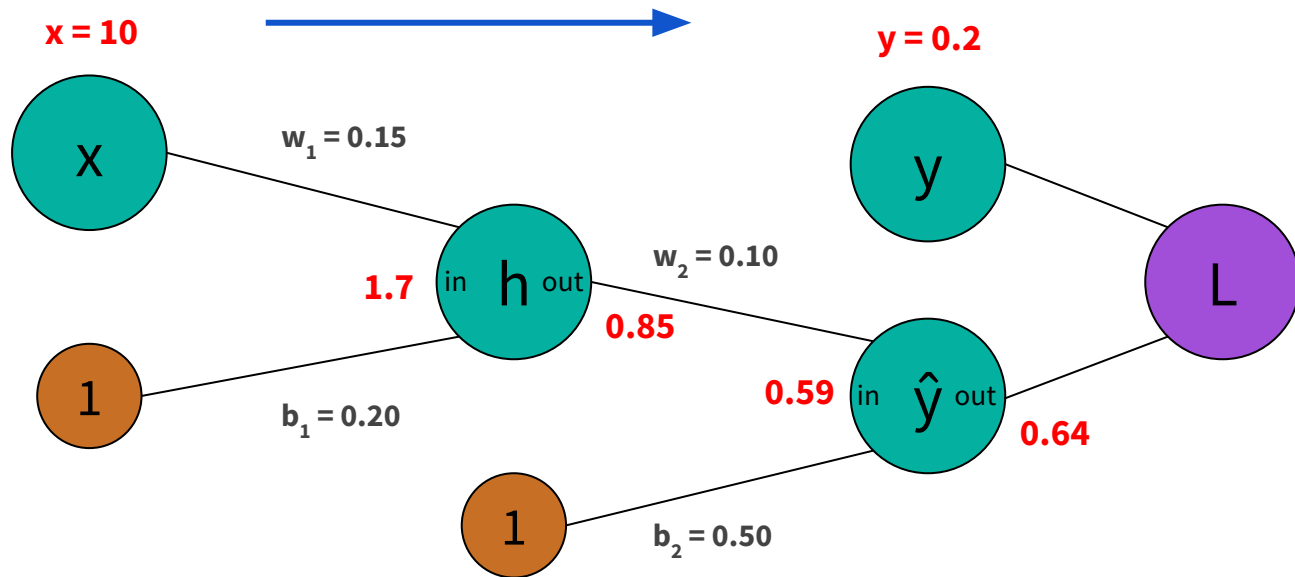
$$\hat{y}_{in} = w_2 h_{out} + b_2 = 0.59$$

$$\hat{y}_{out} = \sigma(\hat{y}_{in}) = 0.64$$

$$L = (y - \hat{y})^2$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

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$$\hat{y}_{in} = 0.10 * 0.85 + 1 * 0.50 = 0.59$$

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Backpropagation - Pasada Forward

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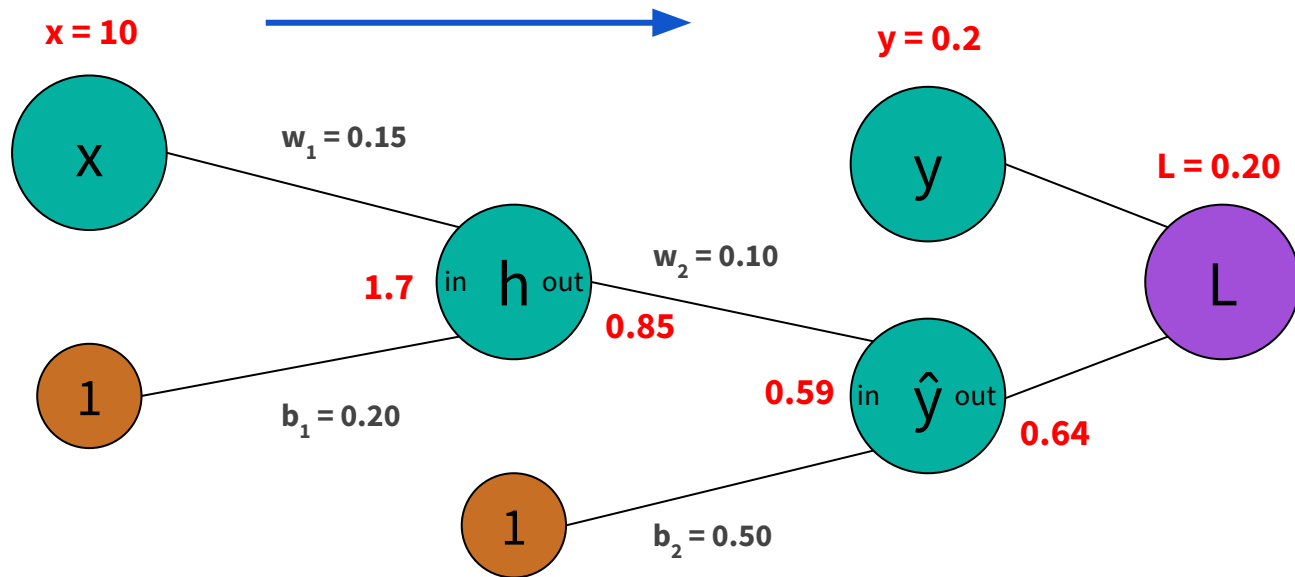
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$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2 = 0.20$$

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$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}}_{\text{out}})^2 = (0.2 - 0.64)^2 = 0.20$$

Backpropagation - Cálculo de derivadas

Gradient Descent nos dice que podemos usar las **derivadas parciales** para buscar un extremo (en este caso un mínimo) de una función, iterativamente:

$$W^{n+1} \leftarrow W^n - \mu \cdot \Delta W^n$$

¿Pero cómo calculamos las derivadas de los pesos?

Usando la regla de la cadena.

¿Y respecto a qué función lo hacemos?

Respecto a la función de pérdida.

Backpropagation - Cálculo de derivadas

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Usando la regla de la cadena.

¿Y respecto a qué función lo hacemos?

Respecto a la función de pérdida.

Queremos calcular $\frac{\partial L}{\partial w_i}$ y $\frac{\partial L}{\partial b_j}$ para todo peso w_i y todo sesgo (bias) b_j

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

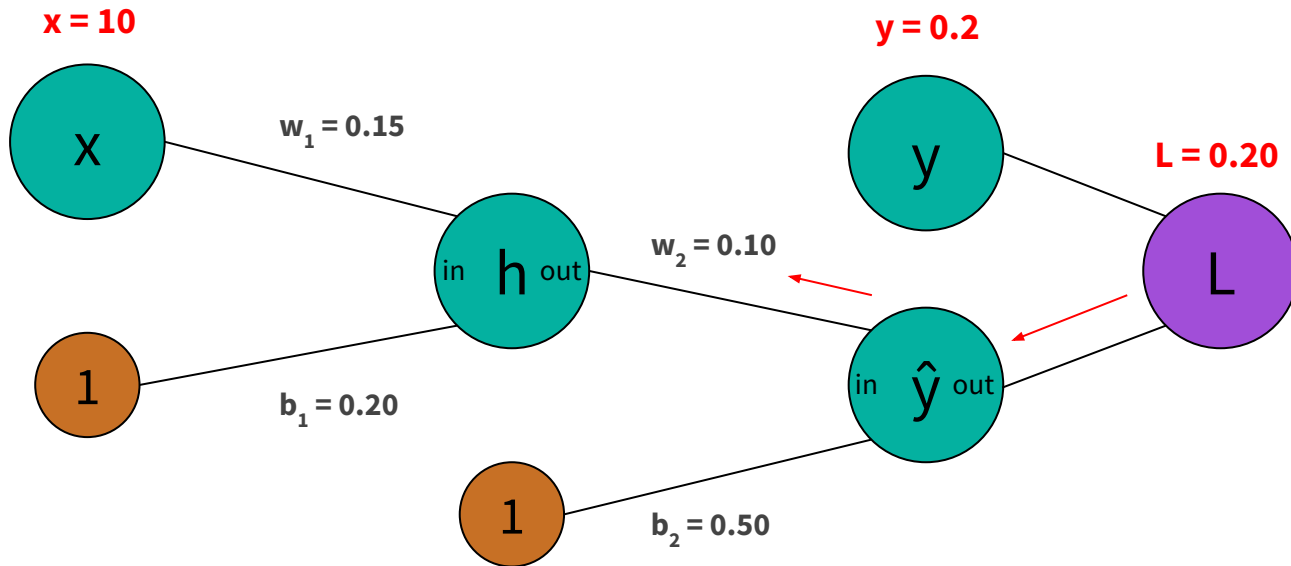
$$\hat{y}_{in} = w_2 h_{out} + b_2 = 0.59$$

$$\hat{y}_{out} = \sigma(\hat{y}_{in}) = 0.64$$

$$L = (y - \hat{y})^2 = 0.20$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

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$$\frac{\partial L}{\partial w_2}$$

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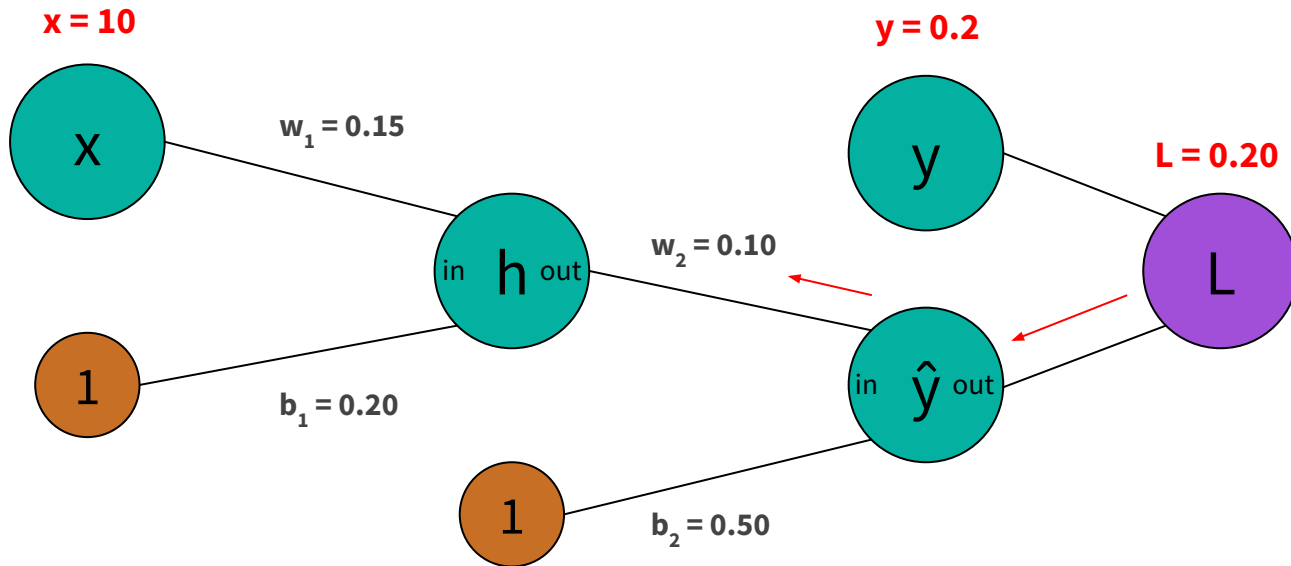
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$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\frac{\partial L}{\partial w_2} = \text{regla de la cadena}$$

Backpropagation - Cálculo de derivadas

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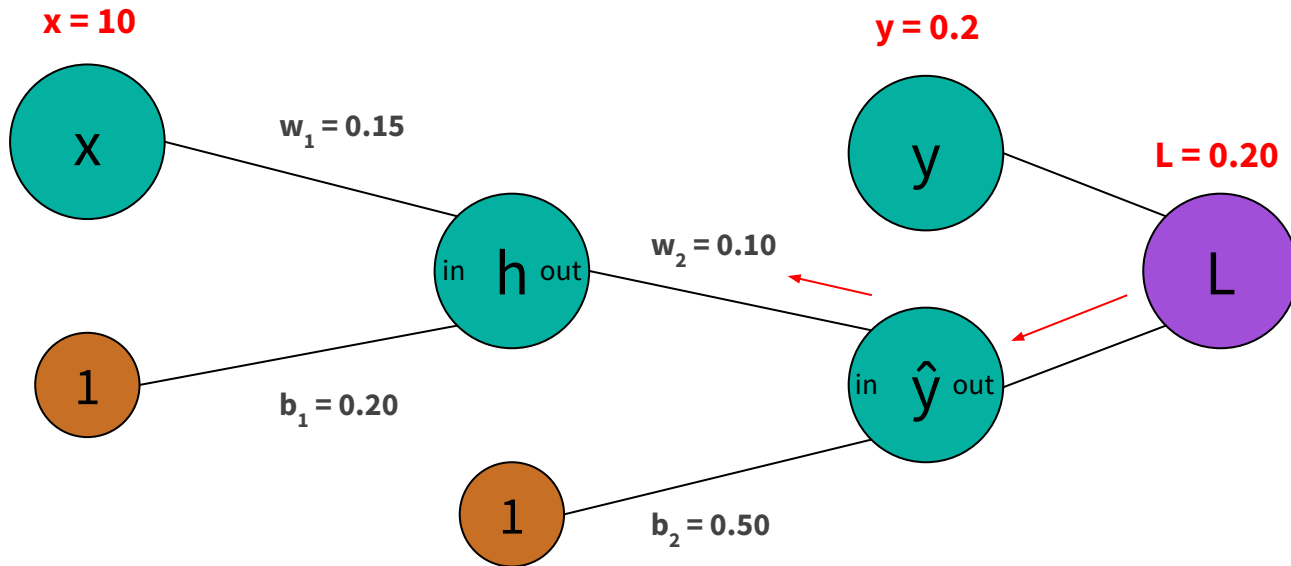
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$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial w_2}$$

Backpropagation - Cálculo de derivadas

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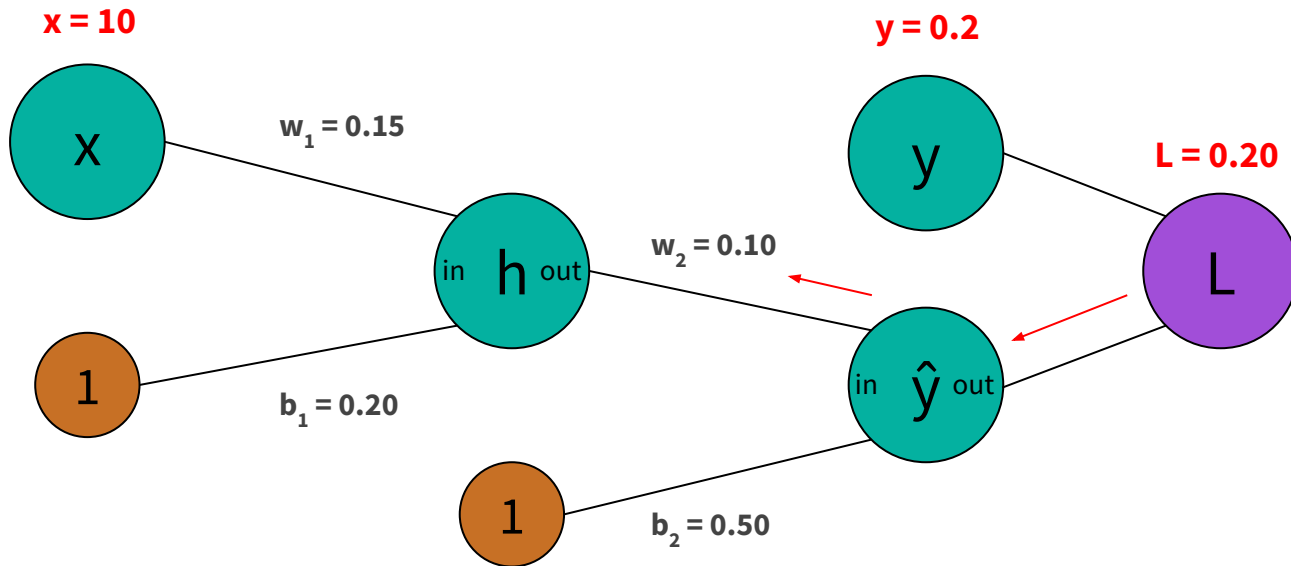
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$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}_{\text{out}}} \frac{\partial \hat{y}_{\text{out}}}{\partial w_2}$$

Backpropagation - Cálculo de derivadas

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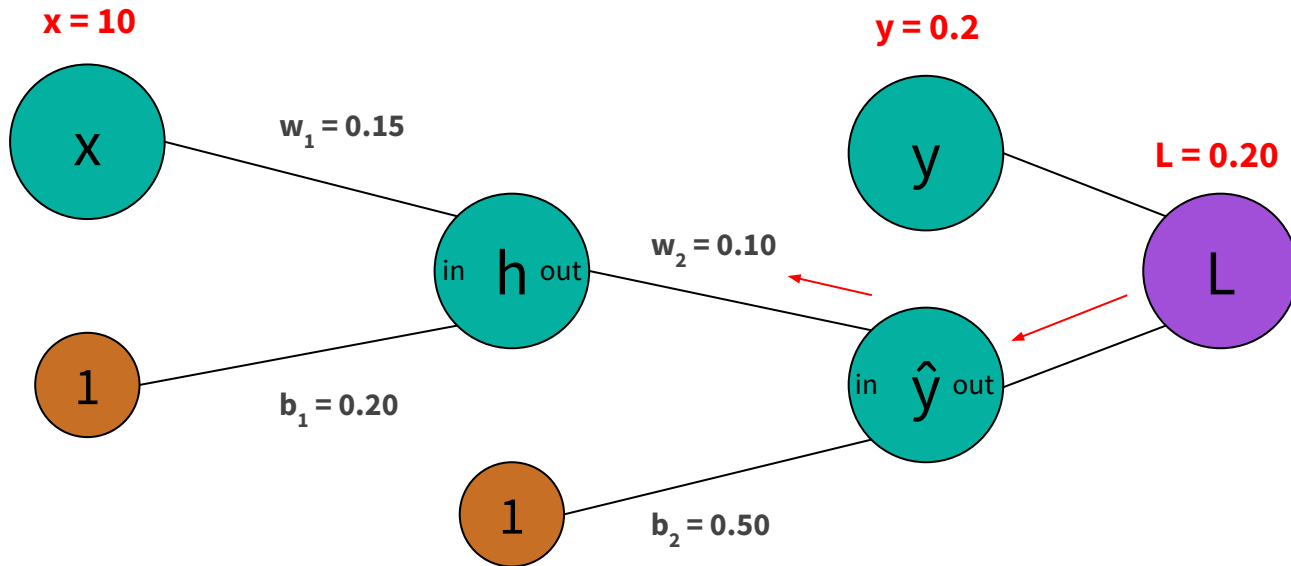
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$$\frac{\partial L}{\partial w_2} = \frac{\partial L}{\partial \hat{y}_{\text{out}}} \frac{\partial \hat{y}_{\text{out}}}{\partial \hat{y}_{\text{in}}} \frac{\partial \hat{y}_{\text{in}}}{\partial w_2}$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

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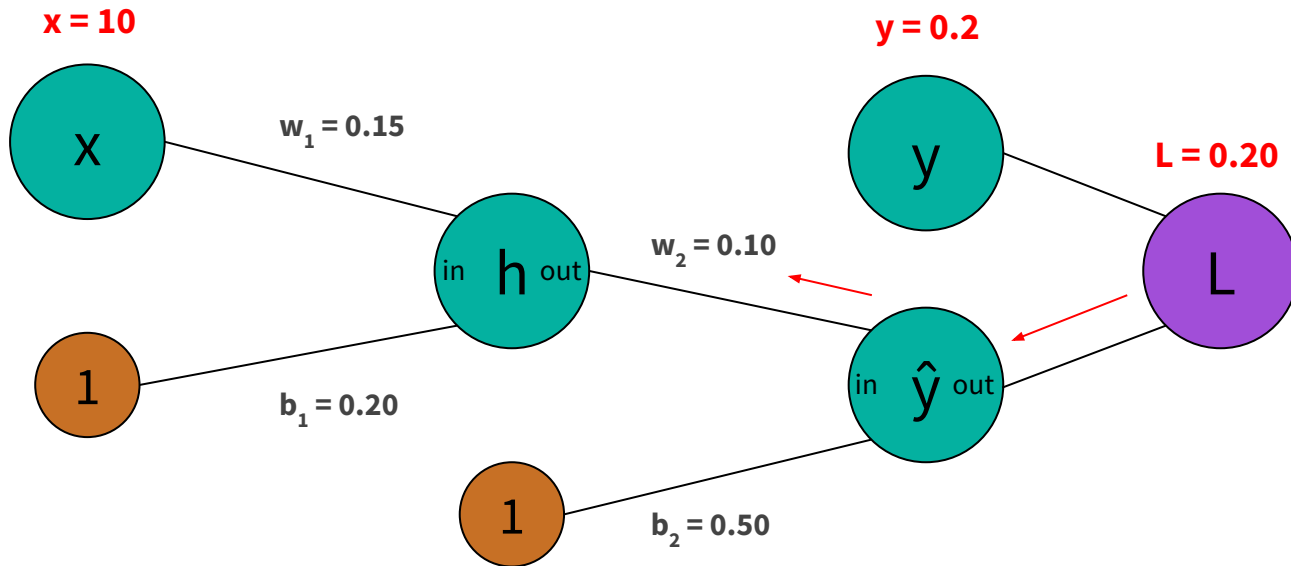
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$$L = (y - \hat{y})^2 = 0.20$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\frac{\partial L}{\partial w_2} = -2(y - \hat{y}_{out}) \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} \frac{\partial \hat{y}_{in}}{\partial w_2}$$

Backpropagation - Cálculo de derivadas

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

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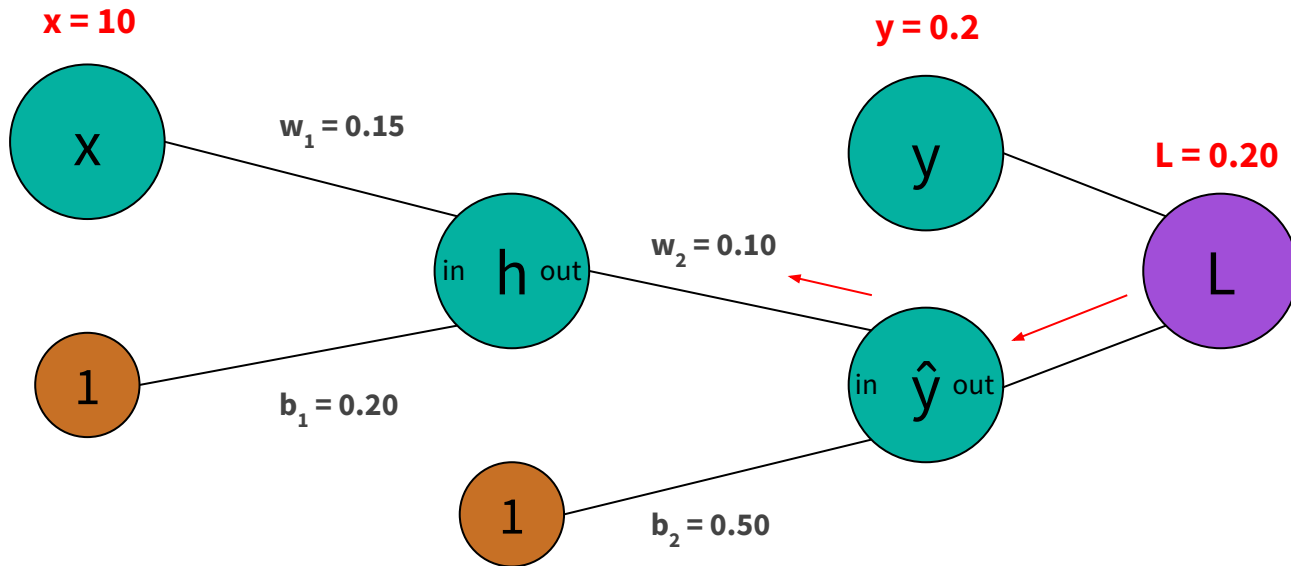
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$$\hat{\mathbf{y}}_{\text{out}} = \sigma(\hat{\mathbf{y}}_{\text{in}}) = 0.64$$

$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2 = 0.20$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

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$$\frac{\partial L}{\partial w_2} = -2(y - \hat{y}_{out}) \sigma(\hat{y}_{in})(1 - \sigma(\hat{y}_{in})) \frac{\partial \hat{y}_{in}}{\partial w_2}$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

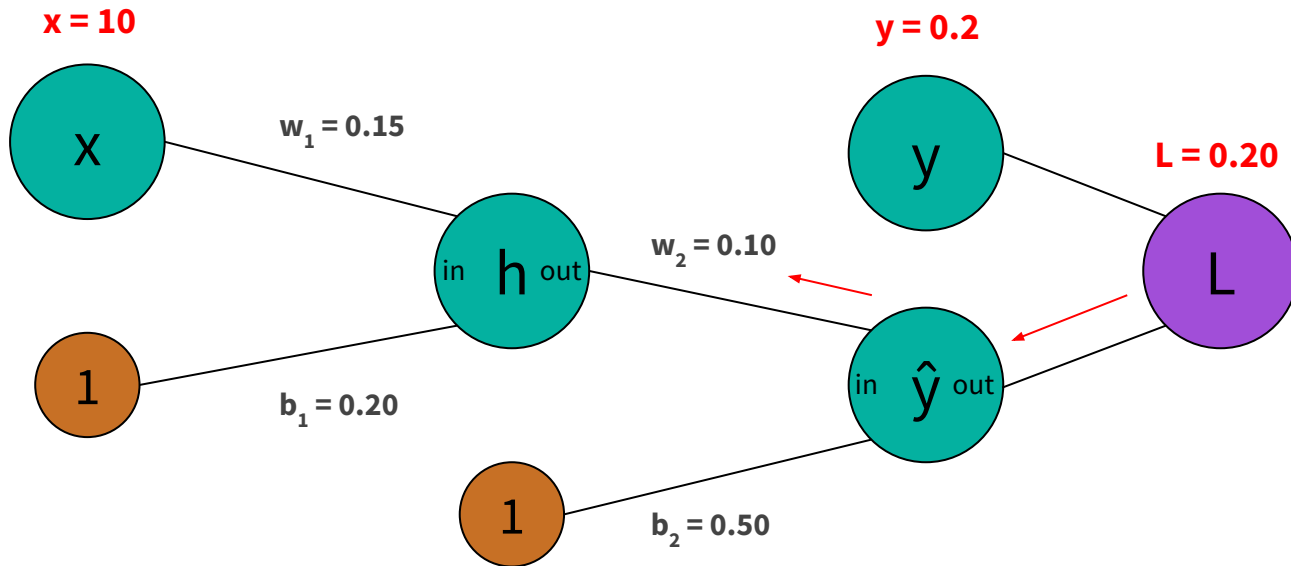
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$$L = (y - \hat{y})^2 = 0.20$$

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$$\frac{\partial L}{\partial w_2} = -2(y - \hat{y}_{out}) \sigma(\hat{y}_{in})(1 - \sigma(\hat{y}_{in})) h_{out}$$

Backpropagation - Cálculo de derivadas

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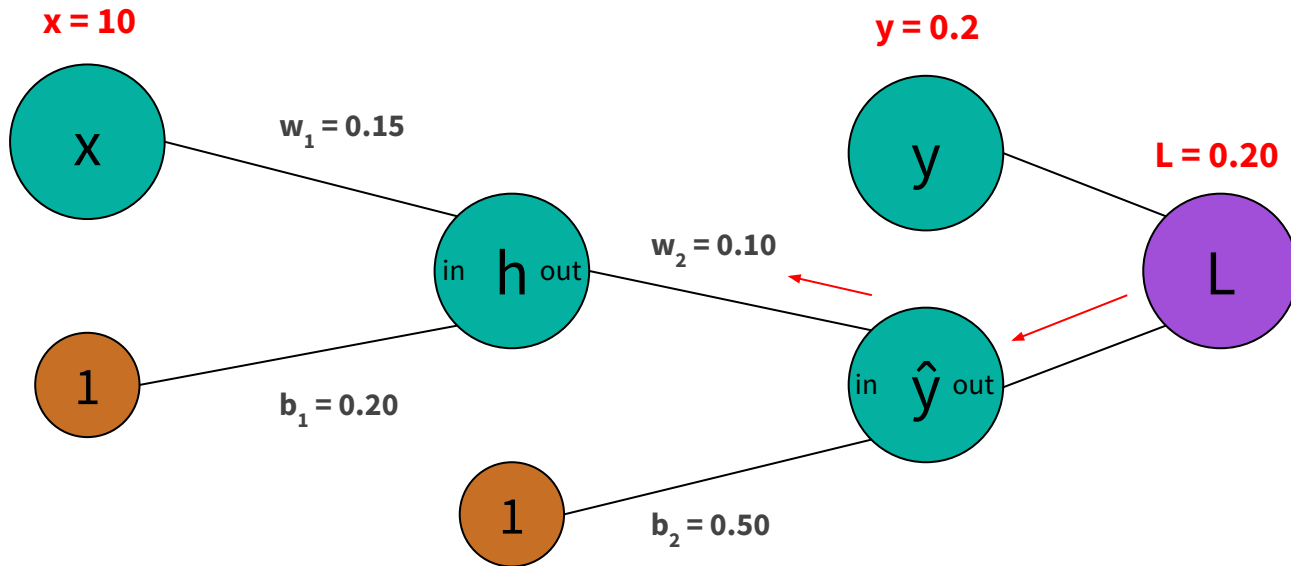
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$$L = (y - \hat{y})^2 = 0.20$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\frac{\partial L}{\partial w_2} = -2(0.2 - 0.64) \sigma(0.59)(1 - \sigma(0.59)) 0.85$$

Backpropagation - Cálculo de derivadas

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

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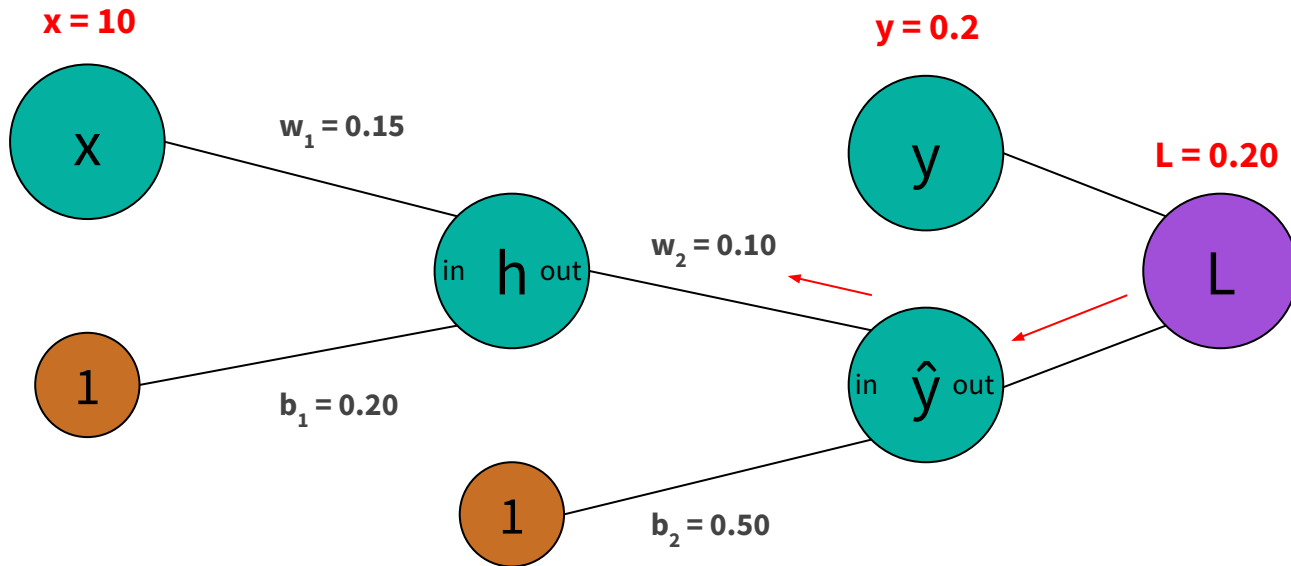
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$$\mathbf{L} = (\mathbf{y} - \hat{\mathbf{y}})^2 = 0.20$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\frac{\partial L}{\partial w_2} = 0.17$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

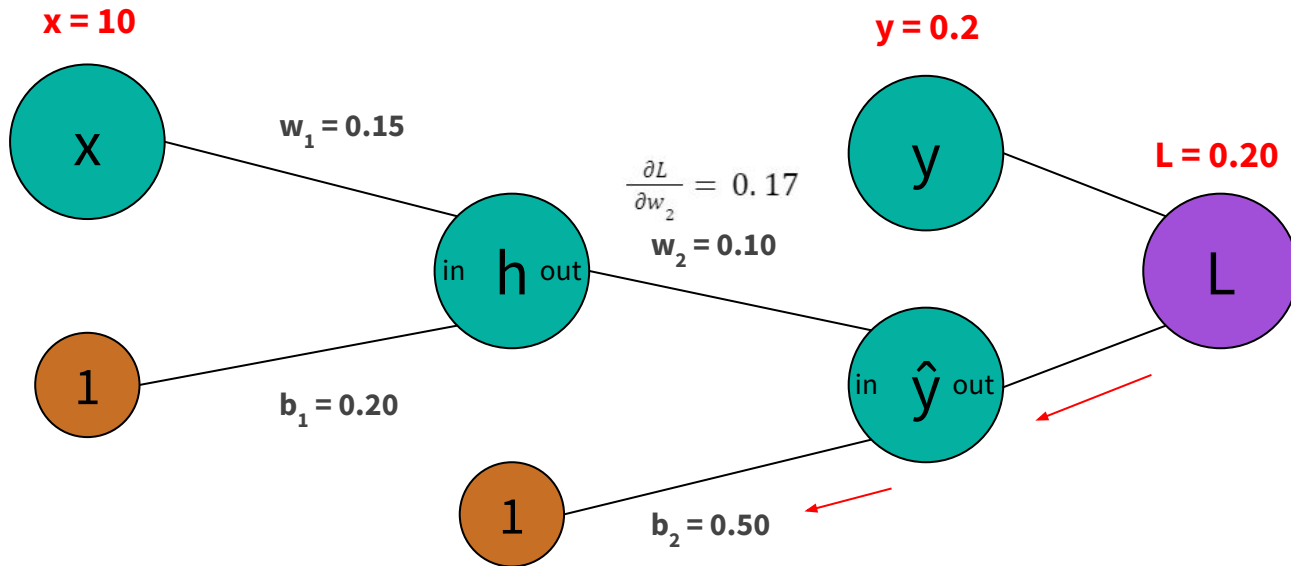
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$$L = (y - \hat{y})^2 = 0.20$$

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$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\frac{\partial L}{\partial b_2} =$$

Backpropagation - Cálculo de derivadas

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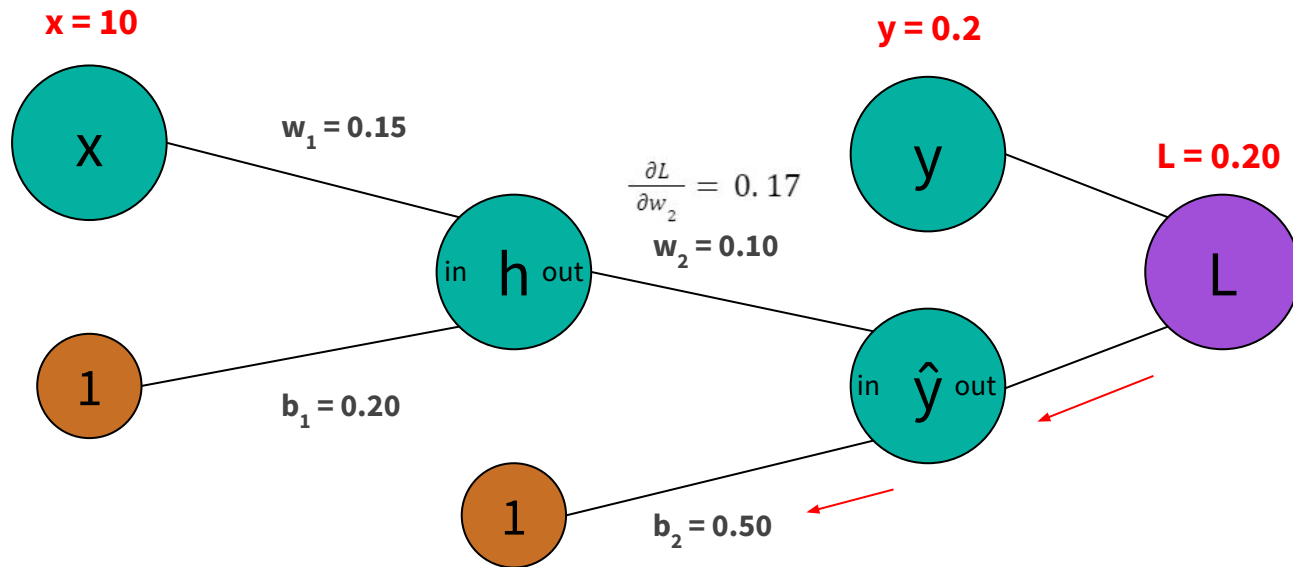
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$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} \frac{\partial \hat{y}_{in}}{\partial b_2}$$

Backpropagation - Cálculo de derivadas

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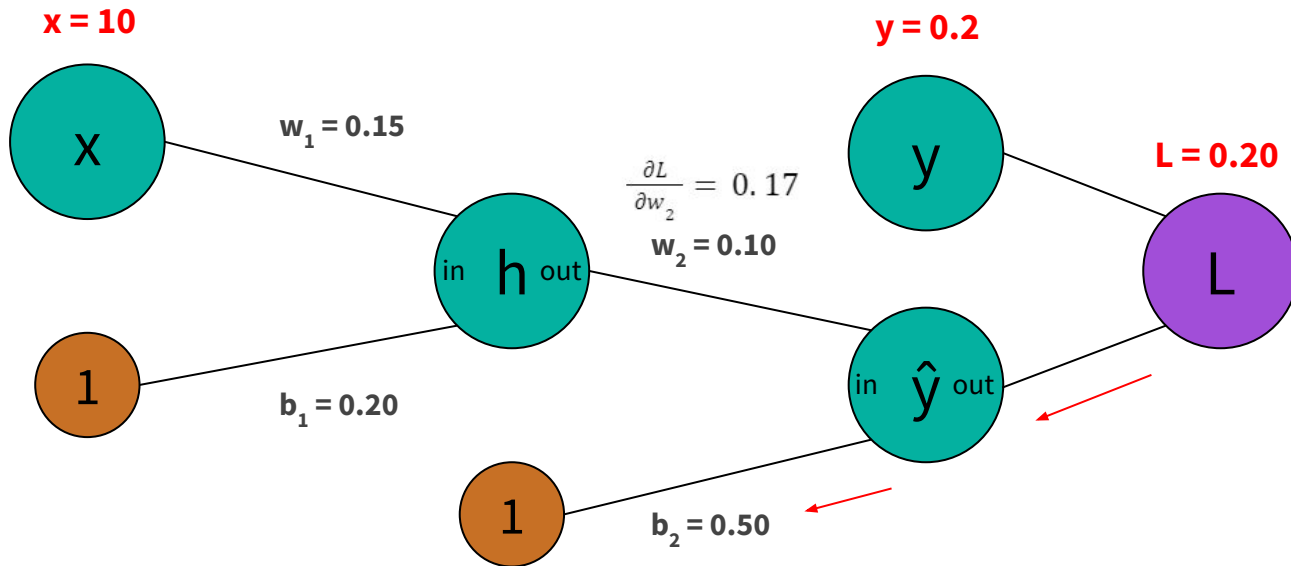
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$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} \frac{\partial \hat{y}_{in}}{\partial b_2}$$

ya tenemos varias!

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

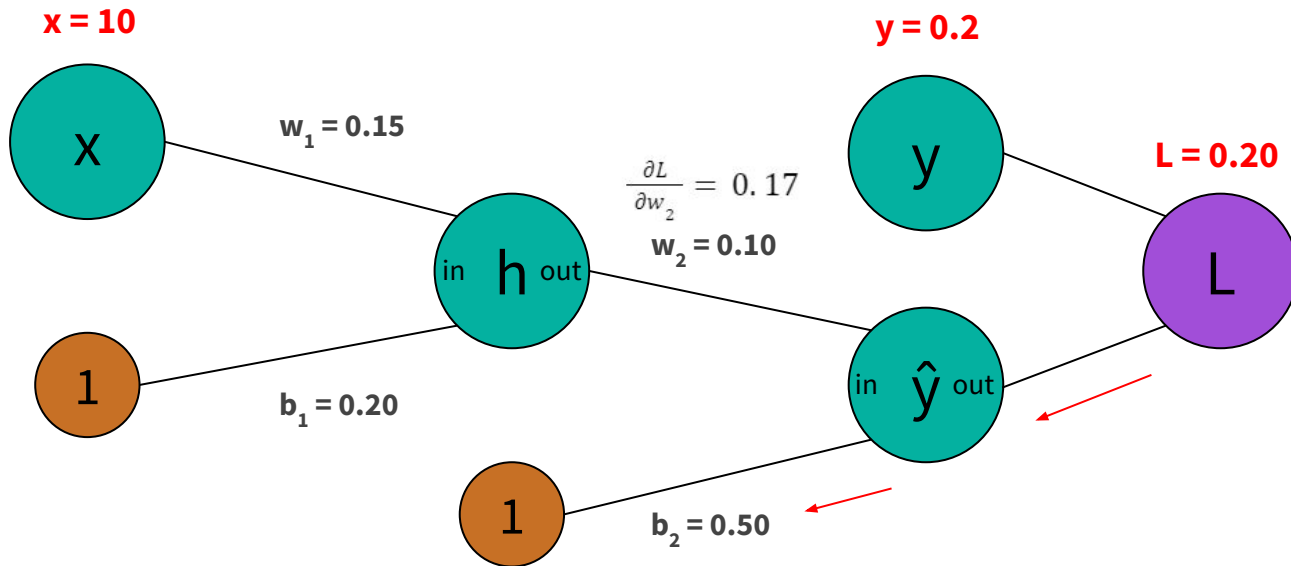
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$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} 1$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

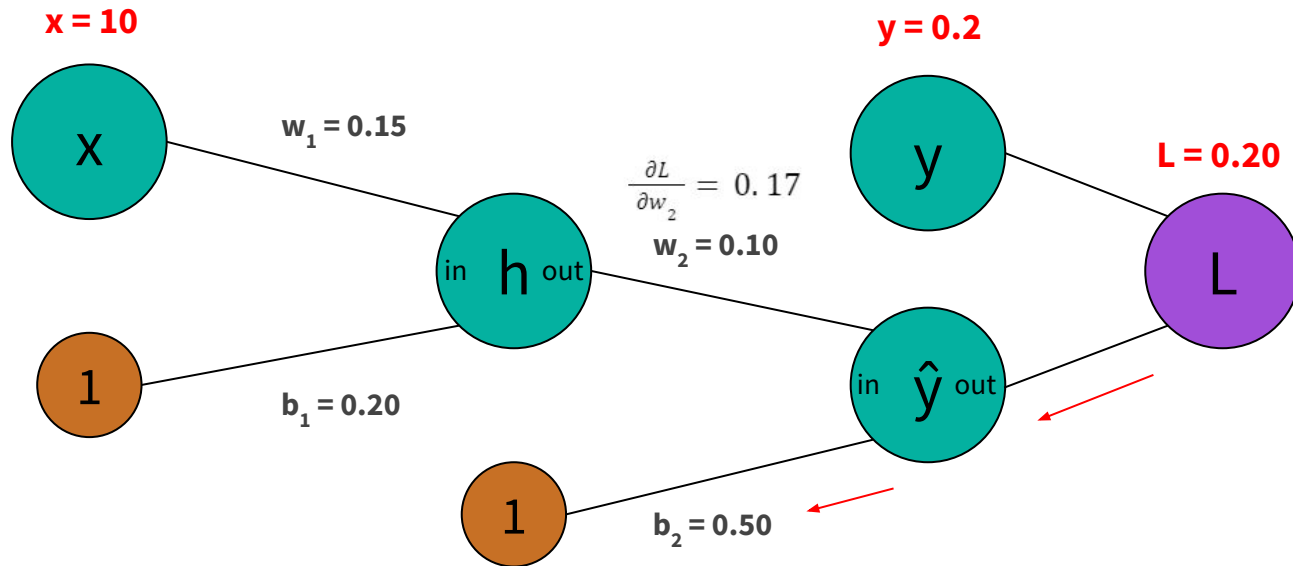
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$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\sigma'(x) = \sigma(x)(1 - \sigma(x))$$



$$\frac{\partial L}{\partial b_2} = -2(y - \hat{y}_{out}) \sigma(\hat{y}_{in})(1 - \sigma(\hat{y}_{in})) 1$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

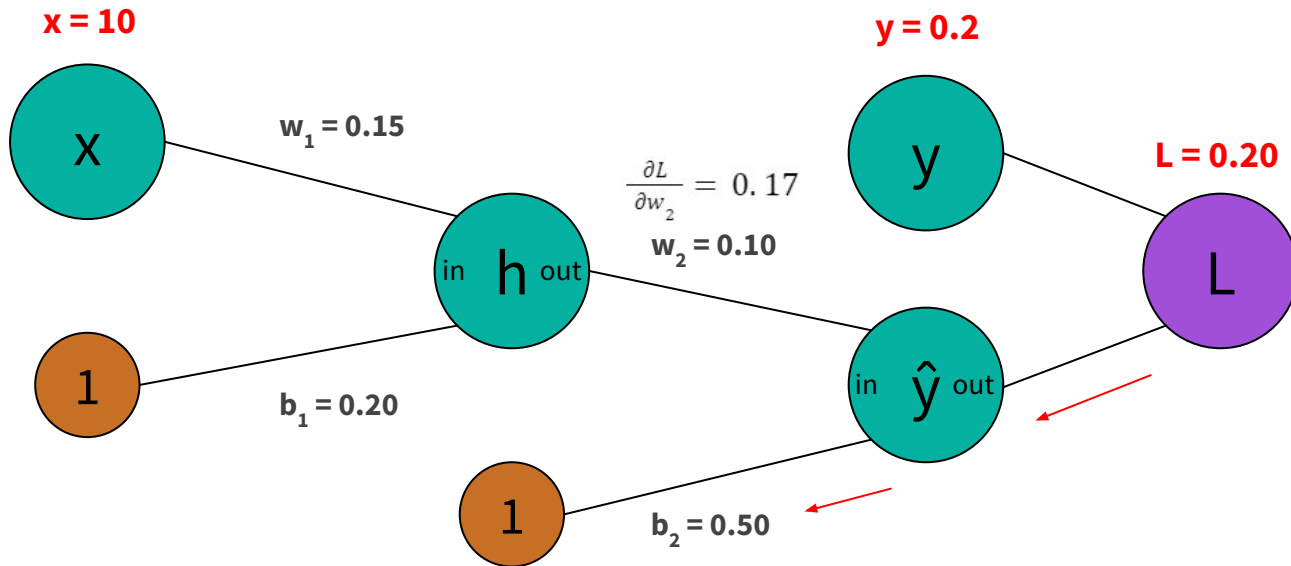
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$$\frac{\partial L}{\partial b_2} = 0.20$$

Backpropagation - Cálculo de derivadas

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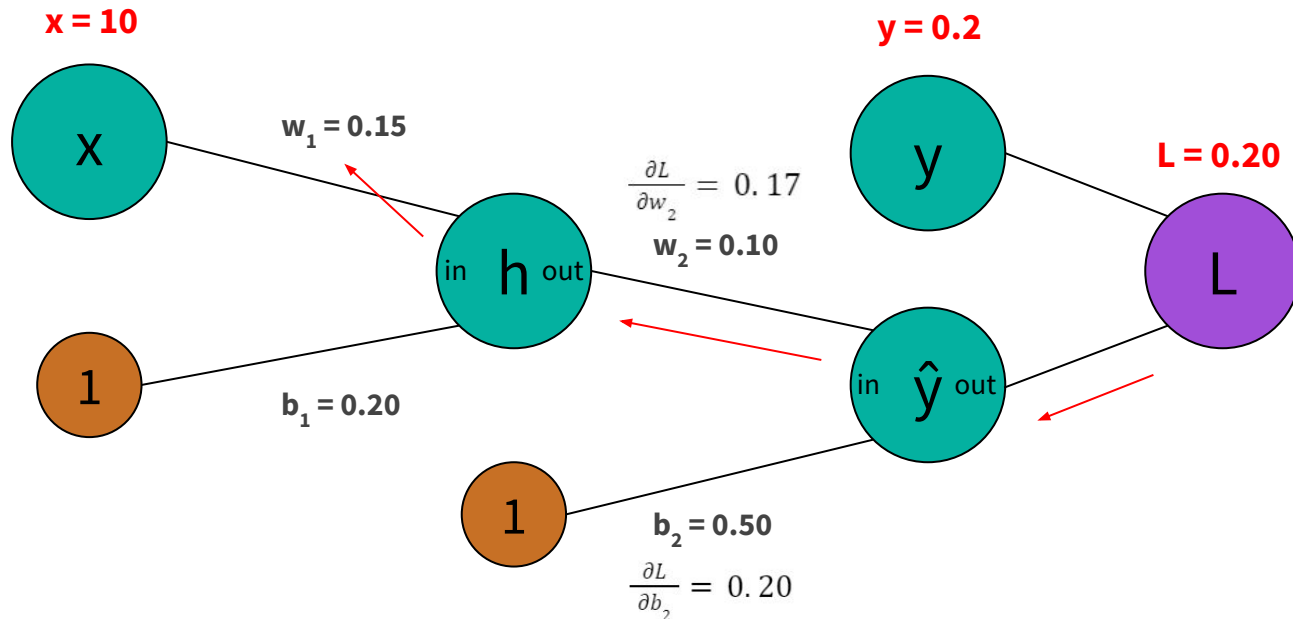
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$$\frac{\partial L}{\partial w_1} =$$

Backpropagation - Cálculo de derivadas

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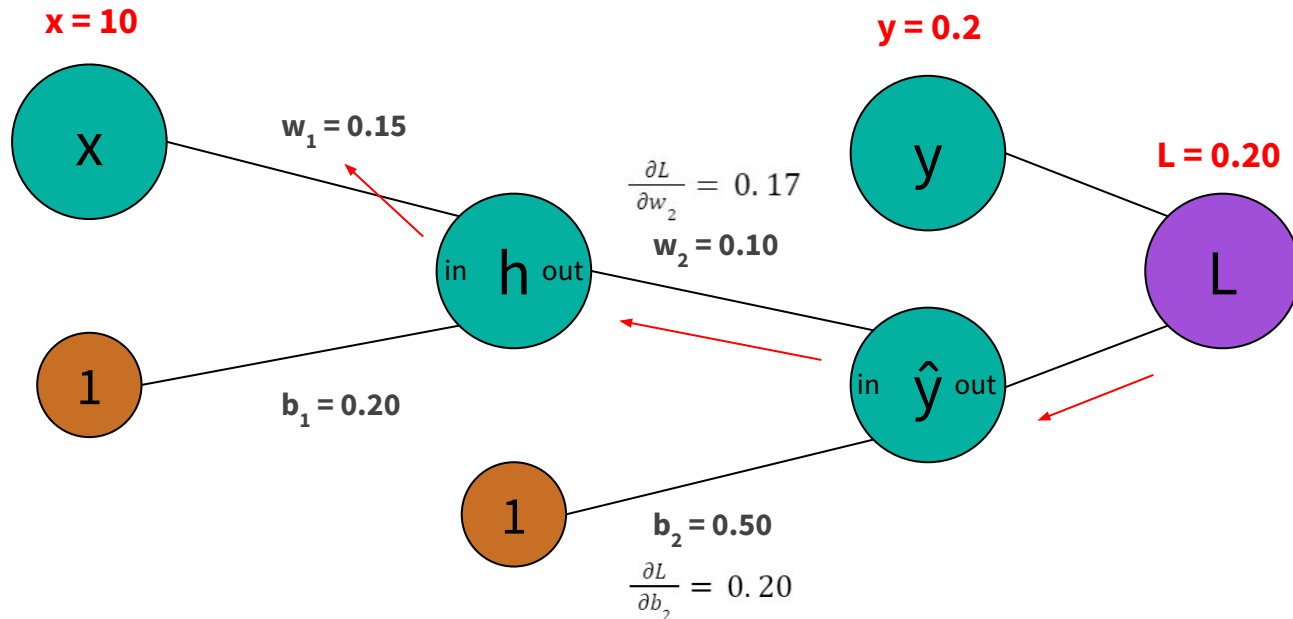
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$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} \frac{\partial \hat{y}_{in}}{\partial h_{out}} \frac{\partial h_{out}}{\partial h_{in}} \frac{\partial h_{in}}{\partial w_1}$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

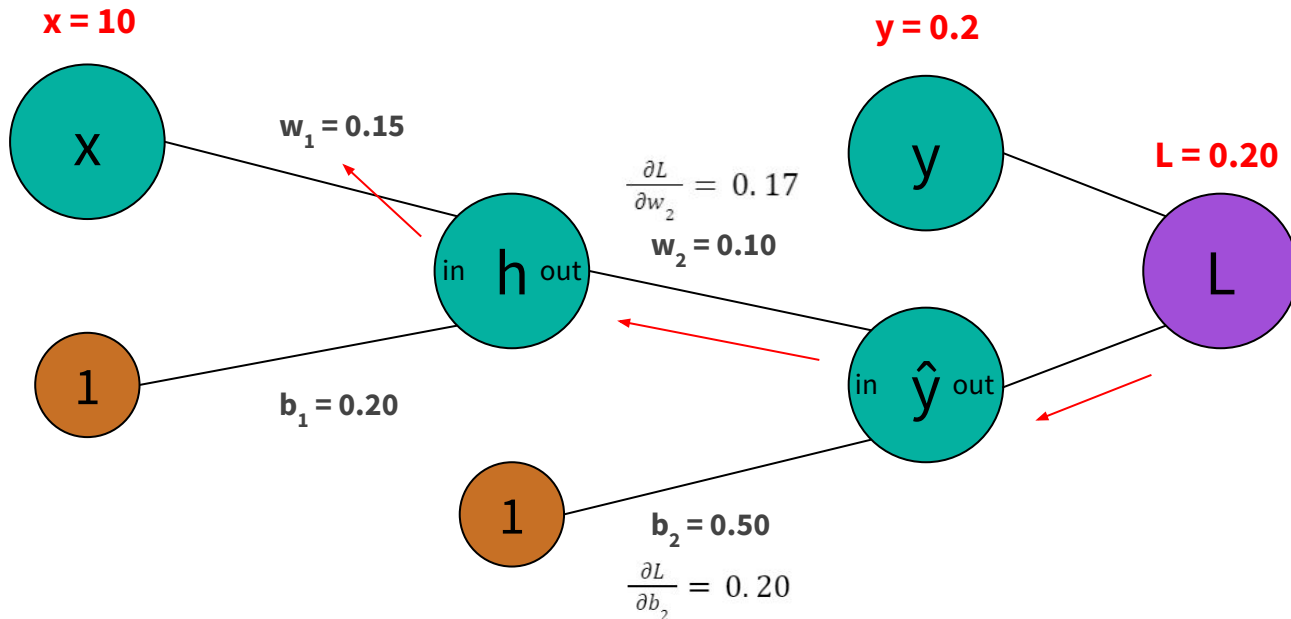
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$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} w_2 \frac{\partial h_{out}}{\partial h_{in}} \frac{\partial h_{in}}{\partial w_1}$$

Backpropagation - Cálculo de derivadas

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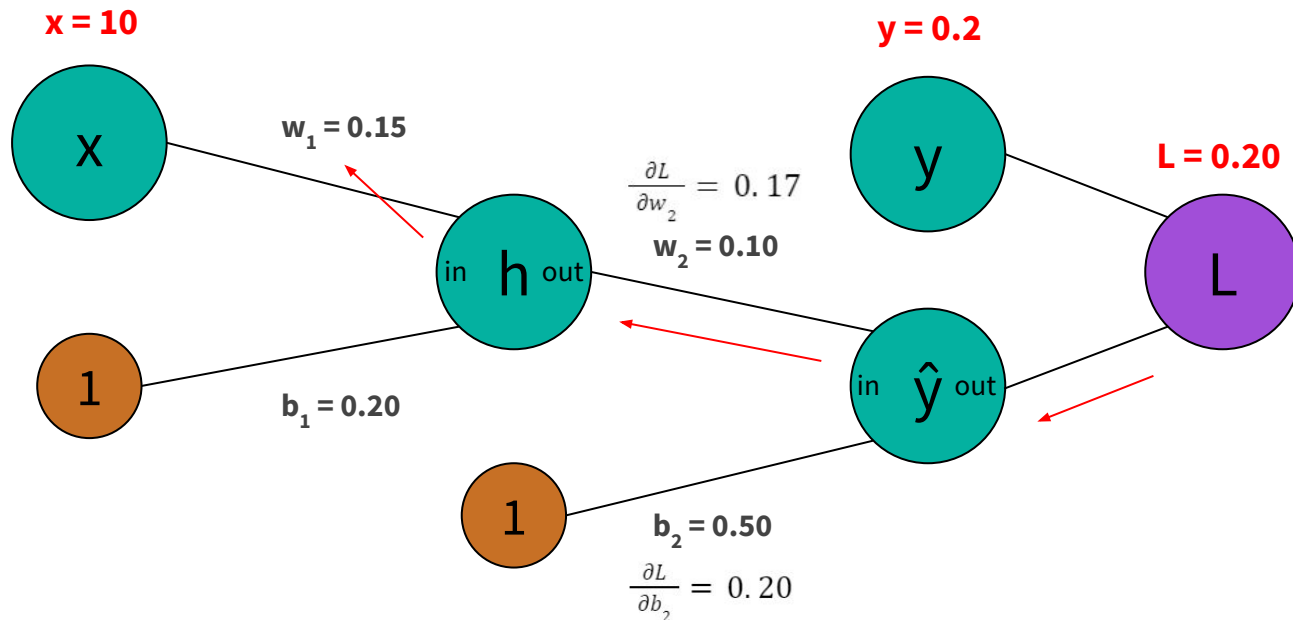
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$$\frac{\partial L}{\partial w_1} = \frac{\partial L}{\partial \hat{y}_{out}} \frac{\partial \hat{y}_{out}}{\partial \hat{y}_{in}} w_2 \frac{\partial h_{out}}{\partial h_{in}} x$$

Backpropagation - Cálculo de derivadas

$$h_{in} = w_1 x + b_1 = 1.7$$

$$h_{out} = \sigma(h_{in}) = 0.85$$

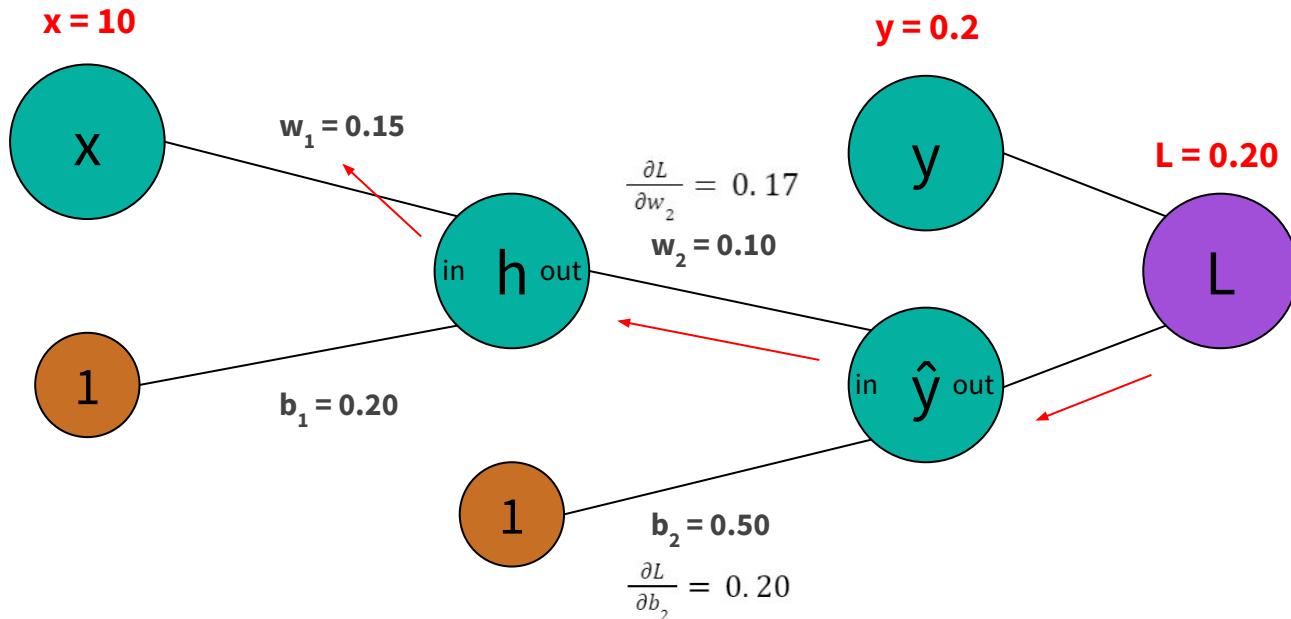
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$$\frac{\partial L}{\partial w_1} = 0.03$$

Backpropagation - Cálculo de derivadas

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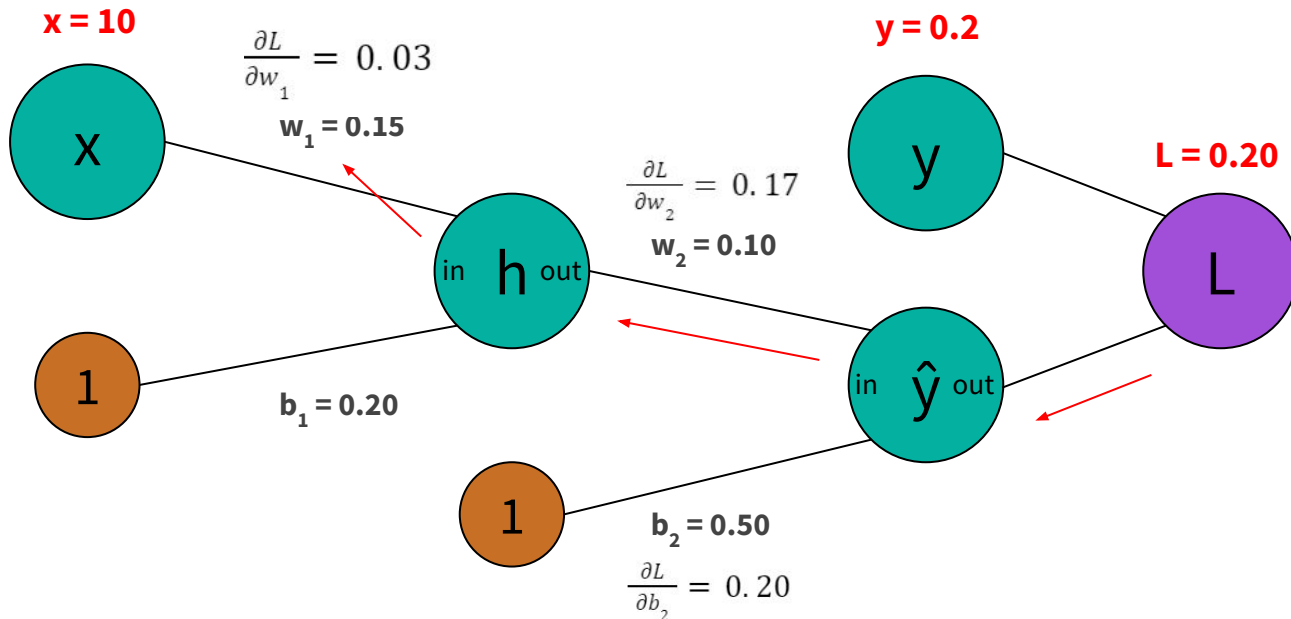
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Backpropagation - Cálculo de derivadas

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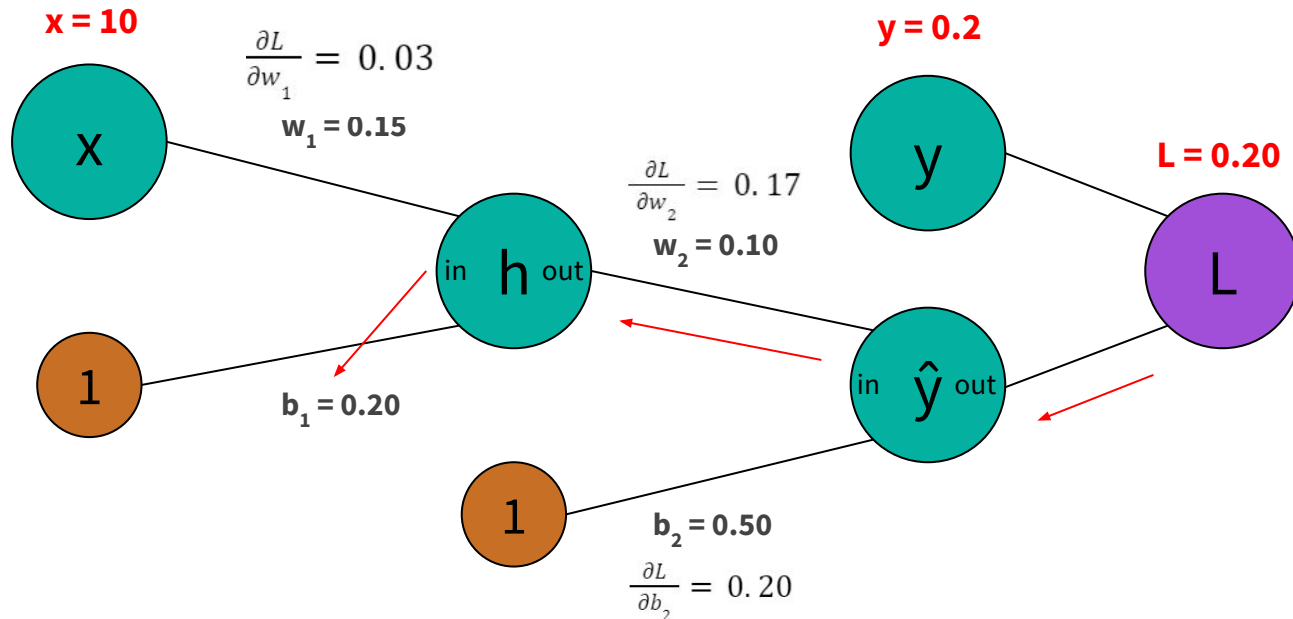
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$$\frac{\partial L}{\partial b_1} =$$

Backpropagation - Cálculo de derivadas

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

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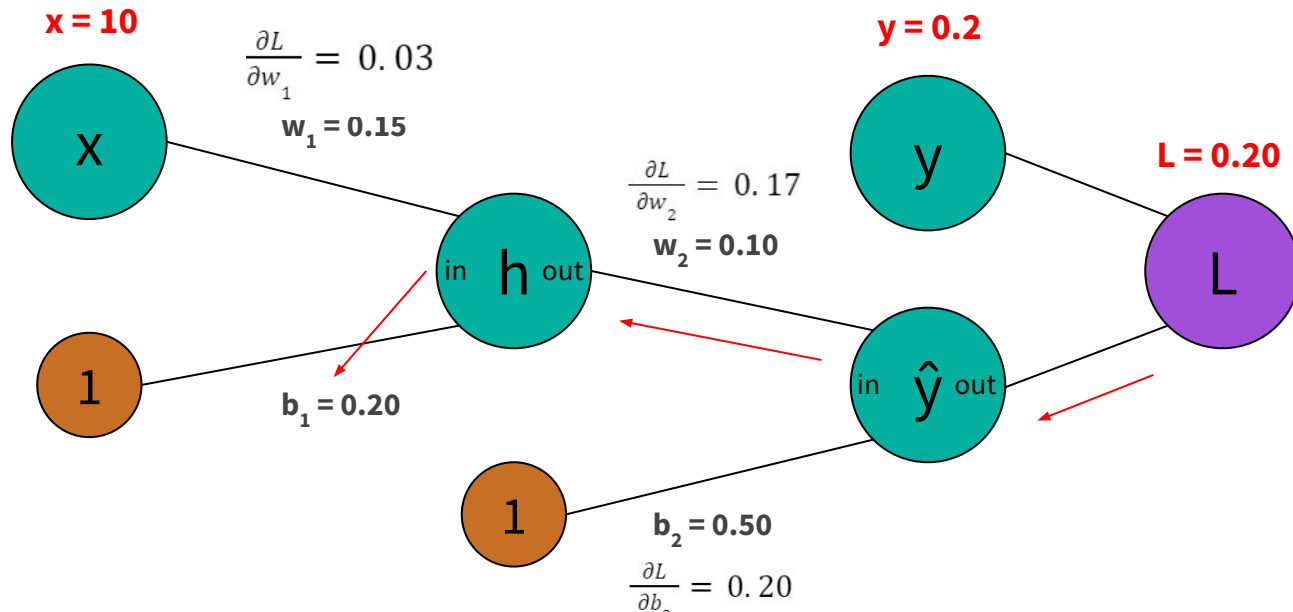
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$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial \hat{\mathbf{y}}_{\text{out}}} \frac{\partial \hat{\mathbf{y}}_{\text{out}}}{\partial \hat{\mathbf{y}}_{\text{in}}} \frac{\partial \hat{\mathbf{y}}_{\text{in}}}{\partial \mathbf{h}_{\text{out}}} \frac{\partial \mathbf{h}_{\text{out}}}{\partial \mathbf{h}_{\text{in}}} \frac{\partial \mathbf{h}_{\text{in}}}{\partial b_1}$$

idéntico al anterior!

Backpropagation - Cálculo de derivadas

$$\mathbf{h}_{\text{in}} = \mathbf{w}_1 \mathbf{x} + \mathbf{b}_1 = 1.7$$

$$\mathbf{h}_{\text{out}} = \sigma(\mathbf{h}_{\text{in}}) = 0.85$$

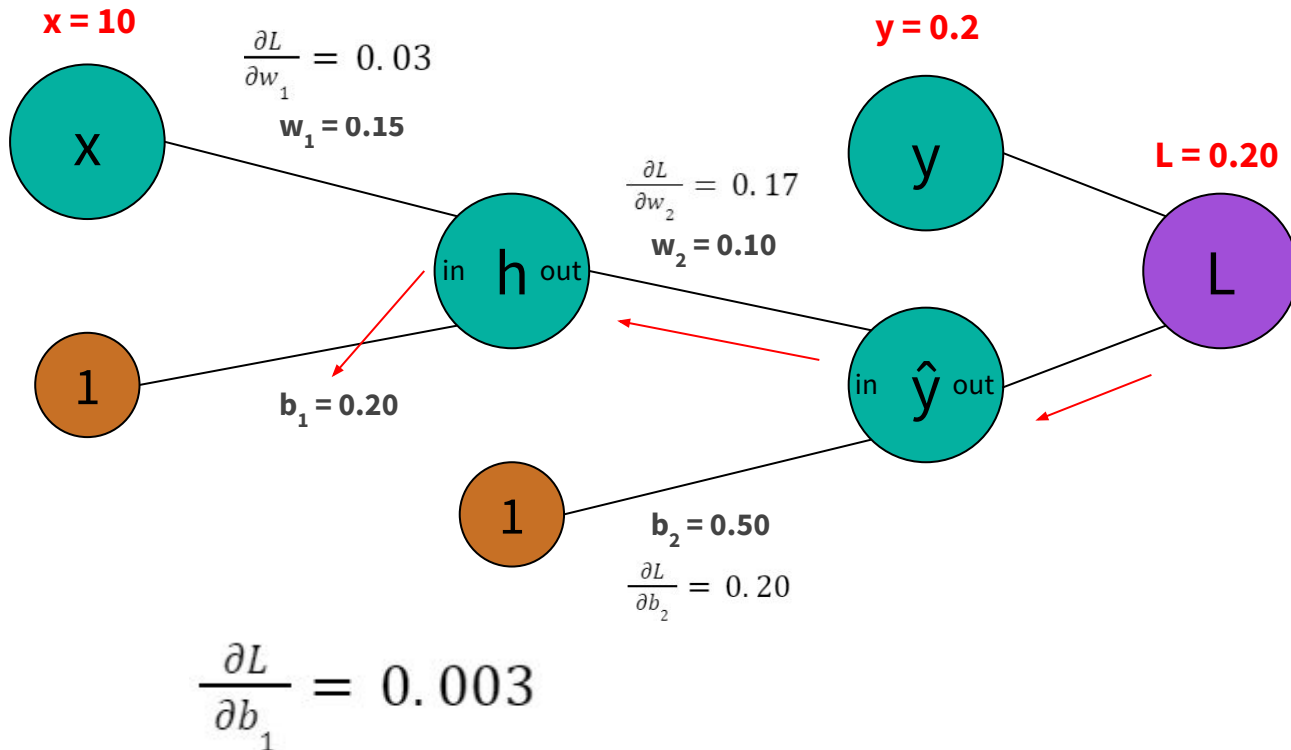
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Backpropagation - Cálculo de derivadas

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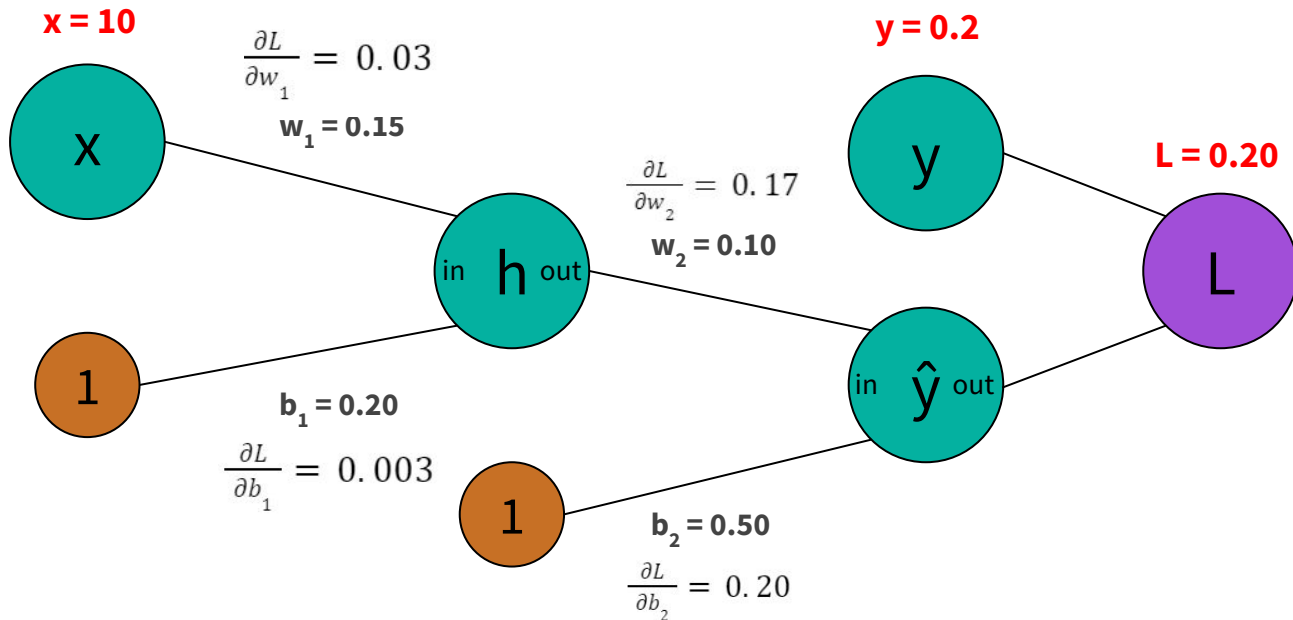
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Backpropagation - Cálculo de derivadas

Después de hacer los cálculos nos quedó:

Backpropagation - Cálculo de derivadas

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$$\frac{\partial L}{\partial w_1} = 0.03$$

$$\frac{\partial L}{\partial w_2} = 0.17$$

$$\frac{\partial L}{\partial b_1} = 0.003$$

$$\frac{\partial L}{\partial b_2} = 0.20$$

Backpropagation - Cálculo de derivadas

Después de hacer los cálculos nos quedó:

$$\frac{\partial L}{\partial w_1} = 0.03$$

$$\frac{\partial L}{\partial w_2} = 0.17$$

$$\frac{\partial L}{\partial b_1} = 0.003$$

$$\frac{\partial L}{\partial b_2} = 0.20$$

¿Qué podemos decir de estos números? ¿Qué representan?

Nos informa qué tanto contribuye un cambio en un peso/sesgo para incrementar el error, respecto a la entrada y salida esperada.

Backpropagation - Actualización de los pesos

Si tomamos un $\mu = 0.5$ (**learning rate**) podemos actualizar los pesos siguiendo la regla:

$$W^{n+1} = W^n - \mu \cdot \Delta W^n$$

Obteniendo los siguientes valores:

$$w_1 = 0.1367$$

$$b_1 = 0.1987$$

$$w_2 = 0.0141$$

$$b_2 = 0.3984$$

Backpropagation - Nuevo Forward Pass

Utilizando los **nuevos parámetros** que calculamos, al hacer un nuevo **forward pass**, nos queda una predicción de **0.6011** ($\hat{y}$) y una pérdida de **0.1609** (L).

Notar que la pérdida es menor a la pérdida inicial : **0.1609** < **0.20**.

Y que $\hat{y}$ está más cerca de y (0.20) : **0.60** vs **0.64**

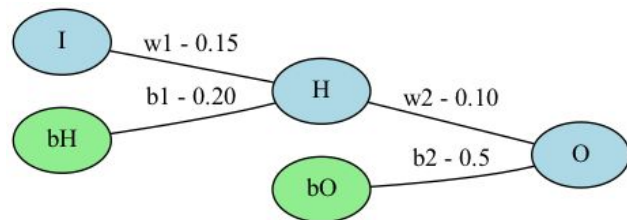
Ejercicio a mano | Chequeo

Pudimos realizar manualmente una pasada forward y otra backward, pero ¿realmente obtuvimos los resultados correctos?

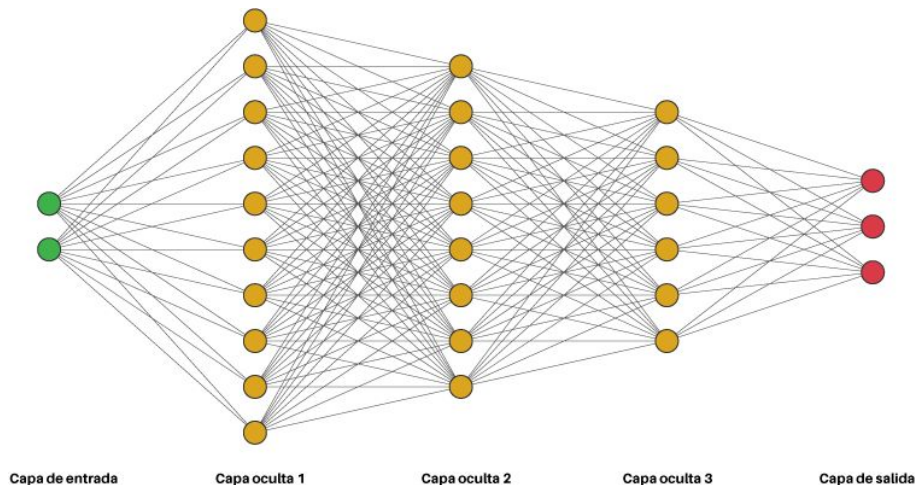
Veámoslo con PyTorch en `td6-p16-d-mano-chequeo.ipynb`.

Ejercicio seleccionado de la guía | Por qué

Hasta ahora, hicimos una sola actualización de los pesos para esta red 🙌



Pero, ¿qué pasa con algo así? 🙌



Ejercicio seleccionado de la guía | Por qué (cont.)

Entonces, queremos una **forma “sistemática” y matricial** de resolver backpropagation, porque:

- Es más **eficiente**.
- Es más **general**.
- Es de fácil **implementación**.

Aclaración: esta implementación no incluye un grafo computacional, como los que usan librerías como PyTorch; es, en cambio, una implementación matricial.

Ejercicio seleccionado de la guía | Por qué (cont.)

Sin embargo, para entender y poder implementar el algoritmo general, es necesario entender **cómo se derivan las ecuaciones generales de backpropagation**.

Esto lo podemos ver resolviendo la cuarta guía de ejercicios (GE4).

Recursos

- Goodfellow, I., Bengio, Y. y Courville, A. (2016). Deep Learning. Disponible [acá](#).
- Hertz, J. A., (2018). *Introduction to the theory of neural computation*.
- [Tutorial en YouTube](#).

Cierre

Hoy vimos y practicamos, entre otras cuestiones,

- cómo podemos resolver manualmente una **pasada forward** y otra **backward** de backpropagation;
- cómo podemos verificar con PyTorch el cambio en los **gradientes** del paso anterior y
- cómo se implementa la **versión generalizada de backpropagation**, basada en **matrices**, resolviendo la GE4.

Pueden darnos *feedback* de la clase [acá](#).